Dangerous Talk[*]

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Abstract

I explore continuous-time partnerships in an organization. Each partner has a productivity-irrelevant hidden type, either normative or rogue. An exogenous public signal process gradually reveals partners' types to each other. The organization incentivizes calling out rogue types, but punishes false accusations. Opportunities to make an accusation or abandon the partnership arrive randomly, and are privately observed — so the partnership's very survival is an endogenous source of information. A symmetric Markov equilibrium exists, and both types are sometimes called out if stakes are high enough.

I fully solve for the unique equilibrium with one-sided incomplete information (say, a stranger entering a purely normative organization), and show how false accusations and partnership dissolutions respond to organizational incentives. In the general case with two-sided incomplete information, the equilibrium is unique when exit opportunities are sufficiently rare. Here, I characterize the equilibrium in terms of fear—exiting because of one's reputation—and openness—collaborating regardless of one's reputation. Lastly, I show that partnerships in this environment are assortative in the long run.

This analysis offers a lens to understand the dynamics of past organizational witch hunts, such as during the Red Scare or the Lavender Scare of the 1950s.

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1. Introduction

Partnerships are formed around shared objectives, but the people who make up these partnerships naturally differ in many other respects, from silly things like favorite food to more relevant topics like political ideologies. These differences, although immaterial to the shared objective that gave rise to the partnership in the first place, may generate tension within the partnership and, in some cases, lead to its dissolution, especially when the differences arise over some prevailing social norm. There are plenty of examples of such cases throughout history, particularly in the U.S. The Red Scare campaign that ran through the late 1940s and 1950s is one such example, where the fear of communist influence turned the suspicion of being a communist into grounds for professional exclusion across many fields, regardless of one's professional record. The most illustrative case of that time is the Hollywood blacklist (Ceplair and Englund, 2003), where studios banned screenwriters and actors from working due to suspicion of communist sympathies. A parallel campaign of the same era was the Lavender Scare, where the focus was the sexual orientation of federal employees, and thousands of them were fired for being suspected of being homosexual (Johnson, 2004).[1]

Today, the scope of what can be perceived as transgressive and provide grounds for exclusion has dramatically expanded with the rise of "cancel culture," in which views at odds with a prevailing norm can cost people their standing among colleagues. As Gibson and Sutherland (2023) document, the share of Americans who do not feel free to speak their minds more than tripled between 1954, at the height of McCarthyism, and 2020, when it reached 46 percent Figure 1.

I present a continuous-time partnership model that captures what these situations share. Each partner privately knows whether she conforms to the prevailing social norm (normative) or departs from it (rogue). Types do not affect the partnership's output, and both partners earn a fixed flow payoff while it lasts, which gives them a reason to stay. Partners learn about each other's types through a pair of public signals that capture their daily interactions. Opportunities to act on this information arrive at random, and each partner observes only her own. When opportunity knocks, a normative type may call out her partner (whistleblow): exposing a rogue earns a reward, while a false accusation is costly to both. A rogue may reveal herself to her partner (scheme): if both are rogue, they carry on as accomplices, in a collaboration worth more than before; otherwise, the

[1]The Lavender Scare was initially driven by concerns that homosexual people were particularly vulnerable to blackmail by foreign powers. However, this form of exclusion outlasted those concerns, as the U.S. military's "Don't Ask, Don't Tell" policy (1994–2011) shows (U.S. Department of Defense, 2024).

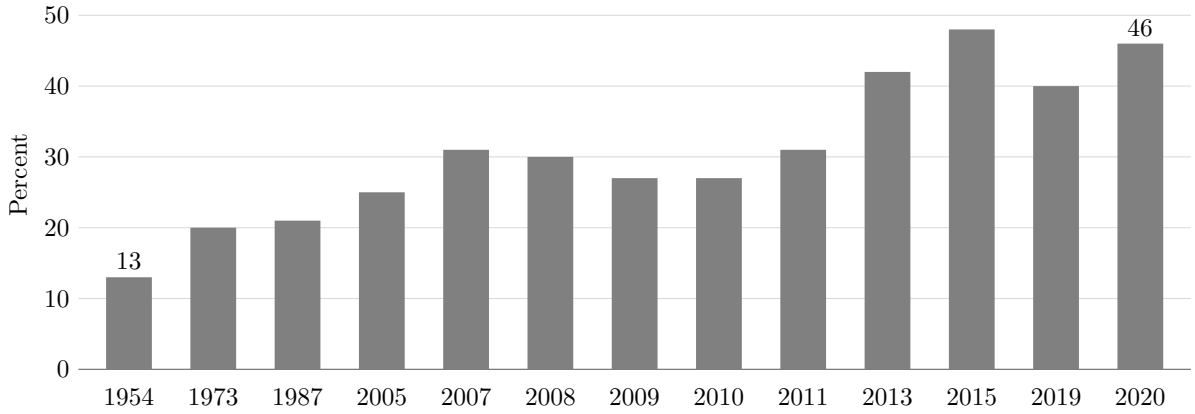


Figure 1: Share of Americans who do not feel free to speak their minds, 1954–2020

Notes: Percentage of respondents who do not feel as free to speak their minds as they used to, from the question "What about you personally? Do you or don't you feel as free to speak your mind as you used to?" Bars are evenly spaced, but the survey years are not. *Source:* Gibson and Sutherland (2023), Figure 1.

scheme exposes the rogue and implicates her partner, imposing a loss on both. Either type may also simply leave (quit) to avoid trouble. A partner may instead pass on the opportunity and leave the partnership undisturbed (collaborate).

Each partner's *reputation* is the posterior belief that the other holds about her type. As public signals change reputations, the two types may have different incentives to continue collaborating, so survival itself can carry information. This information accumulates over time because opportunities to act arrive at random, and continuously because they are private, contributing an additional drift to the reputation dynamics driven by the public signals. Since both the signals and survival are publicly observed, reputations are common knowledge at all times. Yet the two partners assign different laws to the evolution of the same reputation, since only its owner knows her type (Proposition 1).

As in Aumann and Maschler (1995), the analysis begins with one-sided incomplete information and then turns to the two-sided case. In the one-sided case, a *stranger*, whose type is private, works with an *insider* known to be normative, as when someone new joins a purely normative organization. As long as the partnership's output is small relative to both the reward for exposing a rogue and the losses from being called out, the one-sided case has a unique equilibrium, which takes a cutoff form in the stranger's reputation (Proposition 2). The order of these cutoffs depends on what an accusation costs each type (Proposition 3). In the unique equilibrium, the partnership's value falls as the stranger's reputation deteriorates, whatever her type; only partnerships with a normative stranger can last forever, and even these end in a false accusation with positive

probability (Proposition 4). In the two-sided case, where each partner's type is private information, there is a unique symmetric equilibrium when opportunities to act are rare (Proposition 6). Survival is also assortative: mixed partnerships dissolve, while like types stay together longer (Proposition 8).

Related literature. The model studies how partnerships dissolve as beliefs about the partners evolve. In matching models, these beliefs concern match quality (Jovanovic, 1979) or individual productivity (Anderson and Smith, 2010); here, they concern a trait that is irrelevant to joint output. While the partnership lasts, these beliefs respond to two sources of information: the public signals and the partnership's survival. The signals follow the standard Brownian learning framework (Bolton and Harris, 1999; Moscarini and Smith, 2001), in which beliefs evolve as a diffusion. In these models, and in reputation models such as Faingold and Sannikov (2011) and Ekmekci et al. (2022), actions affect learning directly through the signal process. In this model, actions affect learning only indirectly: exit decisions make the partnership's survival informative in equilibrium, and survival adds an endogenous drift on top of the signal-driven diffusion, much as the absence of news does in Keller et al. (2005). There, no news is always bad news. By contrast, survival can move a partner's reputation in either direction, depending on which type wants to exit, and whether that is good or bad news for the other partner depends on her own type. Survival also conveys information in Furusawa and Kawakami (2008), where types differ in their exogenous probabilities of receiving an outside option. Here, types face the same opportunity process, and survival becomes informative through differences in their equilibrium choices about whether to continue collaborating. Exit decisions therefore reveal private information. In such settings, an informed player often randomizes to control how fast she reveals it (Anderson and Smith, 2013; De Angelis et al., 2022). Here, the random arrival of exit opportunities does this work, so the analysis can focus on pure strategies.

Substantively, the model concerns how suspicion erodes the value of productive partnerships when partners care about how others perceive their type. Such concerns shape behavior (Bénabou and Tirole, 2006) and can sustain conformity to a norm (Bernheim, 1994), so people may misrepresent their private views in public (Kuran, 1995). Reputational concerns can also distort advice and actions, destroying information (Morris, 2001; Ely and Välimäki, 2003). In these models, actions affect learning through the signals they generate. Here, the public signal process is exogenous, allowing me to isolate the additional learning generated by endogenous partnership survival.

The paper proceeds as follows. Section 2 presents the model, and Section 3 derives the reputation dynamics and defines equilibrium. Section 4 solves the one-sided case,

and Section 5 turns to the two-sided game. The appendices contain all omitted proofs.

2. Model

PLAYERS AND TYPES. There are two players $i \in \{1, 2\}$.[2] Each has a privately known type $\theta_i \in \{n, r\}$ drawn independently from a common prior $\mathbb{P}(\theta_i = r) = p_0 = 1 - \mathbb{P}(\theta_i = n)$, where $p_0 \in (0, 1)$. Here, n is a *normative* type and r a *rogue* type, with the distinction reflecting whether the player conforms to or departs from a prevailing social norm.

SIGNALS. Time is continuous and indexed by $t \in [0, \infty)$. At $t = 0$, the players enter a partnership, and the game continues until one of them takes a terminal action, described below. Until that time, information about the partners' types is gradually and unintentionally revealed through dialogue and interactions. This information is represented by a pair of public signal processes (Z_t^1, Z_t^2), observed by both players. Specifically, for each $i \in \{1, 2\}$, the process $(Z_t^i)_{t \geq 0}$ evolves according to

$$dZ_t^i = \mu(\theta_i)dt + \sigma dB_t^i,$$

where B^1 and B^2 are independent Brownian motions, independent of the types. The common diffusion coefficient is $\sigma > 0$, and the type-dependent drifts satisfy $\mu(r) > \mu(n)$. The *signal-to-noise ratio* of the process is denoted by $\psi \equiv (\mu(r) - \mu(n))/\sigma$.

OPPORTUNITIES. Players can end the game only at exogenous opportunities, which arrive for player i according to a Poisson process $(N_t^i)_{t \geq 0}$ of intensity $\lambda > 0$, her *opportunity process*. These processes are independent of each other, of the types, and of the Brownian motions (B^1, B^2), so opportunity arrivals by themselves carry no information. Player i's opportunity times are the jump times of N^i,

$$T_k^i := \inf\{t \geq 0 : N_t^i \geq k\}, \qquad k \in \mathbb{N},$$

and $T^i := \{T_k^i : k \in \mathbb{N}\}$ collects them. Since the opportunity processes are independent, the players almost surely never receive opportunities at the same time, so T^1 and T^2 are disjoint.

ACTIONS. When opportunity knocks for player i, at time T_k^i, she chooses an action from a type-dependent set,

$$\mathcal{A}_n := \{c, q, w\}, \qquad \mathcal{A}_r := \{c, q, s\}.$$

[2]Throughout, player i denotes the player under consideration and player $j \neq i$ her partner.

Type-specific Action	Partner's type: Normative	Partner's type: Rogue
whistleblow w	$-F_n,\ -L_n$	$G_n,\ -L_r$
scheme s	$-L_r,\ -L_n$	$G_r,\ G_r$

Table 1: **Terminal Payoffs.** Rows indicate the type-specific action $a \in \{w, s\}$: normative players whistleblow (w) and rogues scheme (s). Columns indicate the partner's type, θ_j. Each cell lists, player i's payoff, $u(a; \theta_j)$, first and her partner's payoff, $u^p(a; \theta_j)$, second. All parameters are positive: G_θ and L_θ denote a type-θ player's gain and loss, respectively, and F_n is the cost of making a false accusation.

She may collaborate (c) by passing on the opportunity. This choice leaves no trace: since opportunities are private, her partner cannot tell whether she passed on an opportunity or had none. She may instead take a *terminal action*, which both players observe and which ends the game. Either type may quit (q); in addition, a normative type may whistleblow (w), that is, call out her partner, and a rogue type may scheme (s), that is, come out to her partner. A player who quits or whistleblows dissolves the partnership. A rogue who schemes dissolves the partnership if her partner is normative; if her partner is rogue, the two carry on as accomplices. Since the model concerns only collaboration under uncertainty about types, the payoff from a terminal action summarizes, in present value, everything that follows.

Payoffs. Each player accrues a flow payoff $\pi > 0$ while the game continues and receives a terminal payoff when a terminal action is taken. The terminal payoff depends on this action, the partner's type, and which player takes the action. Table 1 reports the terminal payoffs when player i takes the type-specific action $a \in \{w, s\}$ (whistleblow or scheme): her own payoff $u(a; \theta_j)$ first and her partner's $u^p(a; \theta_j)$ second. The common terminal action, quit, pays both types the same amount, normalized to zero, irrespective of who takes it: $u(q; \cdot) = u^p(q; \cdot) = 0$.

Letting τ denote the time at which the game ends and a the terminal action taken, player i's discounted payoff is

$$\int_0^\tau e^{-\rho t}\pi\, dt + e^{-\rho\tau}\left[u(a;\theta_j)\,\mathbb{1}\{\tau \in T^i\} + u^p(a;\theta_i)\,\mathbb{1}\{\tau \in T^j\}\right].$$

The payoff structure reflects the central tension of the game. Calling out a player (whistleblow) triggers an investigation, which reveals her type and imposes on her a loss L_θ. For a rogue type, the accusation brings a severe backlash, while for a normative type it is a nuisance until the investigation clears her name. For the accuser, calling someone

out earns a reward G_n when it correctly targets a rogue, but a false accusation against a normative type carries a cost F_n. Coming out as a rogue (scheme) is beneficial between rogues, who each earn G_r, but detrimental when the partner is normative: the rogue loses L_r and her partner L_n. The stakes of accusations and schemes are large relative to the value of the partnership, so each player must be wary of what others think about her type. Together, these considerations motivate the parameter restrictions $L_r > L_n$ and, for each type $\theta \in \{n, r\}$, $\rho G_\theta > \pi$ and $\lambda L_\theta > \pi$. The last two give the tension its bite: a player sure that her partner is a rogue acts on it, and one who expects her partner to act against her would rather leave than wait. The analysis also excludes knife-edge parameter values.[3]

3. Analysis

3.1. Information and Beliefs

Players learn from two public sources of information. First, they observe the signals (Z^1, Z^2), which generate the filtration $\{\mathcal{Z}_t\}_{t\geq 0}$.[4] Second, they observe whether the game continues. That is, at time t, they observe the *survival event* $\{\tau > t\}$. Since a player's decision to take a terminal action may depend on her type, the absence of such actions can convey information beyond what the public signals contain. Accordingly, I define player i's public posterior as[5]

$$P_t^i := \mathbb{P}(\theta_i = r \mid \mathcal{Z}_t, \{\tau > t\}),$$

the probability that she is rogue conditional on the signal history and continued play. For the analysis, it is convenient to express beliefs in log-odds form. I define player i's *reputation* as

$$X_t^i := \log\left(\frac{P_t^i}{1 - P_t^i}\right), \quad \text{with inverse} \quad P_t^i = p(X_t^i), \quad \text{where} \quad p(x) := \frac{1}{1 + e^{-x}}.$$

Since $p(\cdot)$ is a strictly increasing bijection from $\mathbb{R}$ onto $(0, 1)$, reputation X_t^i and posterior belief P_t^i carry the same information. The reputation profile is a measurable function of public information and is therefore common knowledge while the game con-

[3] Specifically, $\pi + \rho F_n \neq \lambda(L_n - F_n)$.

[4] Let $\mathcal{Z}_t^0 := \sigma(Z_s^1, Z_s^2 : 0 \leq s \leq t)$ be the raw filtration generated by the public signals. I use the usual augmentation, defined by $\mathcal{Z}_t := \bigcap_{u>t}(\mathcal{Z}_u^0 \vee \mathcal{N})$, where $\mathcal{N}$ consists of all subsets of $\mathbb{P}$-null events in the underlying probability space. The resulting filtration is complete and right-continuous.

[5] For a sigma-algebra $\mathcal{F}$ and events E_1, E_2, I use $\mathbb{P}(E_1 \mid \mathcal{F}, E_2)$ as shorthand for $\mathbb{P}(E_1 \cap E_2 \mid \mathcal{F})/\mathbb{P}(E_2 \mid \mathcal{F})$, wherever $\mathbb{P}(E_2 \mid \mathcal{F}) > 0$.

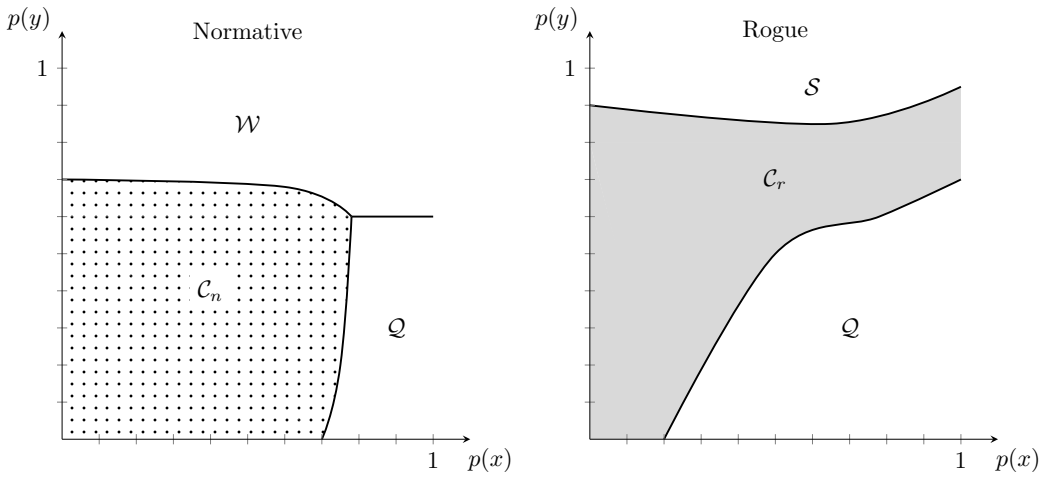


Figure 2: **Decision Rules**. The figure illustrates a strategy $\alpha = (\alpha_n, \alpha_r)$. Each type's decision rule partitions the state space into a collaboration region and stopping regions.

tinues. Moreover, the players' types remain conditionally independent given the public signal history and continued play: the signal likelihoods factor across players, as does the survival likelihood conditional on the signals. Consequently, (X_t^1, X_t^2) determines the joint posterior distribution of (θ_1, θ_2) and summarizes the public information about types. Motivated by this representation, I focus on stationary Markov strategies, under which each player's action at an opportunity depends only on her type and the current reputation profile.

NOTATION. The variables x and y denote realizations of X_t^i and X_t^j, respectively, and θ' denotes the type opposite to θ. The analysis is conducted in log-odds space, while figures are reported in belief space to ease exposition.

3.2. Strategies and State Dynamics

Player i acts only when opportunity knocks, selecting an available action based on her privately known type θ_i and the current reputation profile (x, y). Formally, a pure stationary Markov strategy is a pair of measurable mappings $\alpha = (\alpha_n, \alpha_r)$, where $\alpha_\theta : \mathbb{R}^2 \to \mathcal{A}_\theta$ is the *decision rule* for type θ, specifying the action taken at state (x, y). Each decision rule partitions the state space according to the prescribed action. For each type θ, the *collaboration region*

$$\mathcal{C}_\theta(\alpha) := \{(x, y) \in \mathbb{R}^2 : \alpha_\theta(x, y) = c\}$$

consists of the states at which that type passes on an opportunity, allowing the game to continue. Analogously, $\mathcal{Q}_\theta(\alpha)$ denotes the quitting region for type θ, while $\mathcal{W}(\alpha)$ and

$\mathcal{S}(\alpha)$ denote the normative type's whistleblowing region and the rogue type's scheming region, respectively. An example of the resulting partitions is shown in Figure 2.

Collaboration regions need not coincide across types: at a given state, a strategy may prescribe collaboration for one type and a terminal action for the other. These differences make continued play informative about a player's type. To identify the relevant states in which the survival event $\{\tau > t\}$ is informative, define the *informational region* of type θ as

$$\mathcal{I}_\theta(\alpha) := \mathcal{C}_\theta(\alpha) \setminus \mathcal{C}_{\theta'}(\alpha).$$

This region consists of the states at which type θ collaborates but type θ' would take a terminal action. While the state remains in this region, the game continues under type θ' only if no opportunity arrives. Since opportunities arrive at rate λ, the probability of no arrival falls as time passes, so the survival of the game increasingly favors type θ. Because opportunities are private, this evidence accumulates continuously until the game ends or the state leaves the informational regions. Measured in log-odds, the evidence grows at the constant rate λ: continued play moves the reputation toward rogue in $\mathcal{I}_r(\alpha)$, toward normative in $\mathcal{I}_n(\alpha)$, and is uninformative elsewhere. The *survival drift* measures this effect at each state:

$$\Lambda(x, y; \alpha) := \lambda \left(\mathbb{1}\{(x, y) \in \mathcal{I}_r(\alpha)\} - \mathbb{1}\{(x, y) \in \mathcal{I}_n(\alpha)\} \right).$$

The next proposition gives the laws of motion of reputations, taking into account both the information conveyed by the public signals and what continued play reveals given the strategies.

Proposition 1 (Reputation Processes)**.** *Fix a strategy profile (α, β). From player i's perspective, the reputations (X_t^i, X_t^j) evolve according to*

$$dX_t^i = \left[\Lambda(X_t^i, X_t^j; \alpha) + \frac{\psi^2}{2} \left(\mathbb{1}\{\theta_i = r\} - \mathbb{1}\{\theta_i = n\} \right) \right] dt + \psi dB_t^i, \tag{1}$$

$$dX_t^j = \left[\Lambda(X_t^j, X_t^i; \beta) + \frac{\psi^2}{2} \tanh\left(X_t^j / 2 \right) \right] dt + \psi \, d\widetilde{B}_t^j, \tag{2}$$

for some Brownian motion $\widetilde{B}^j$ with respect to the public information.

The asymmetry between the two laws of motion reflects private knowledge rather than disagreement about the public state. Player i knows θ_i and therefore conditions the signal component of X_t^i on her true type, which gives rise to the indicator functions. By contrast, she does not know θ_j, so she averages the signal drift of X_t^j over the current

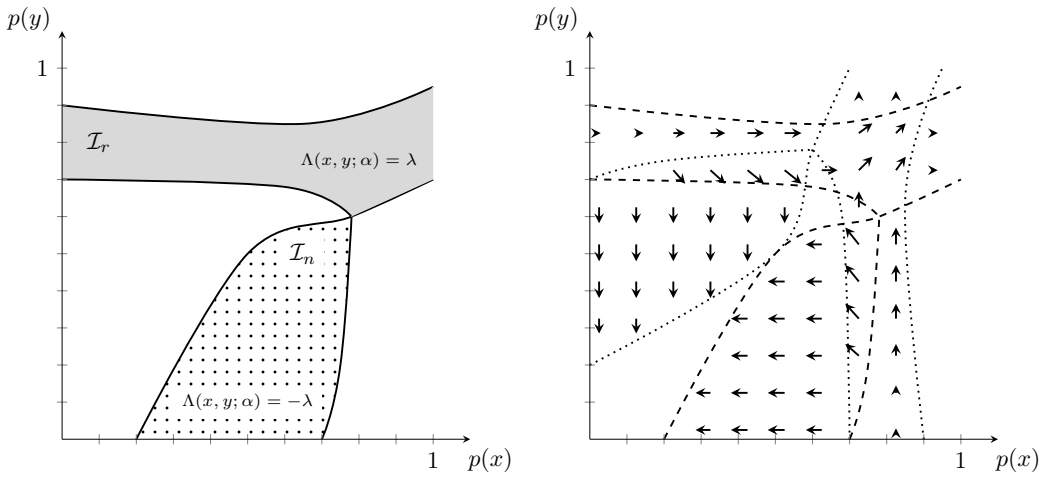


Figure 3: **Survival Drift.** Left: informational regions and values of Λ for the strategy α shown in Figure 2. Right: survival-induced drift field in belief space under the symmetric strategy profile (α, α). Arrows lengths are proportional to the drift magnitude in probability coordinates, accounting for the conversion from log odds to probabilities.

posterior, which yields the term $\tanh(X_t^j/2)$. The noise term changes accordingly: $\widetilde{B}^j$ is the innovation in player j's signal, its surprise relative to that average. Both players nonetheless observe the same reputation profile at every instant; private information only gives them different expectations about how it will evolve.

The survival drift, however, enters both laws of motion in the same way. Since the strategy profile is common knowledge in equilibrium, both players draw the same inference from continued play. Through the survival drift, players' actions shape the evolution of reputations even though they do not affect the signal process. Figure 3 illustrates the survival drift induced by an example strategy.

3.3. Equilibrium Concept

I focus on symmetric stationary Markov perfect equilibria (MPE). In such an equilibrium, the common strategy α plays two roles: it prescribes behavior through the decision rules α_n and α_r, and it determines how continued play updates reputations through the survival drift $\Lambda(\cdot; \alpha)$. Equilibrium requires consistency between these roles: each type's decision rule must be optimal given the partner's behavior and the reputation updating induced by α.

To make the equilibrium condition explicit, fix a conjectured strategy α, common to both players, and allow player i to deviate to an arbitrary measurable decision rule $\varphi : \mathbb{R}^2 \to \mathcal{A}_{\theta_i}$. Public reputations continue to be computed from observed histories using the conjectured profile (α, α), while φ determines player i's actions at opportunity

times. The deviation induces the stopping time and terminal action:

$$\tau_i(\varphi;\alpha) := \inf\{T_k^i : \varphi(X^i_{T_k^i}, X^j_{T_k^i}) \in \mathcal{A}_{\theta_i} \setminus \{c\}\}, \quad a_i(\varphi;\alpha) := \varphi(X^i_{\tau_i(\varphi;\alpha)}, X^j_{\tau_i(\varphi;\alpha)}).$$

The game ends at $\tau = \min\{\tau_i(\varphi;\alpha), \tau_j(\alpha_{\theta_j};\alpha)\}$, and the realized terminal action a^τ is $a_i(\varphi;\alpha)$ if player i stops first and $a_j(\alpha_{\theta_j};\alpha)$ if player j does. The expected payoff of player i's deviation is

$$J_{\theta_i}(x,y;\varphi,\alpha) := \mathbb{E}_{x,y}\left[\int_0^\tau e^{-\rho t}\pi\, dt + e^{-\rho\tau}\left(u(a^\tau;\theta_j)\,\mathbb{1}\{\tau_i=\tau\} + u^p(a^\tau;\theta_i)\,\mathbb{1}\{\tau_j=\tau\}\right)\middle|\,\theta_i\right],$$

where $\mathbb{E}_{x,y}$ denotes expectation given the initial state (x,y), with reputations evolving as in Proposition 1 for the profile (α,α).

A symmetric MPE is a triple (α^*,τ^*,V^*) of type-indexed pairs, $\alpha^* = (\alpha_n^*,\alpha_r^*)$, $\tau^* = (\tau_n^*,\tau_r^*)$, and $V^* = (V_n^*,V_r^*)$, such that for every type $\theta \in \{n,r\}$ and every state $(x,y) \in \mathbb{R}^2$,

$$\alpha_\theta^* \in \operatorname*{argmax}_\varphi J_\theta(x,y;\varphi,\alpha^*), \qquad \tau_\theta^* = \tau_i(\alpha_\theta^*;\alpha^*), \qquad V_\theta^*(x,y) = J_\theta(x,y;\alpha_\theta^*,\alpha^*).$$

The first condition requires each type's equilibrium decision rule to be a best response to the partner's behavior and the belief-updating rule induced by α^*. The remaining conditions define the associated stopping times and continuation values.

4. One-Sided Learning

To fix ideas, this section considers a one-sided learning environment. A newcomer, *the stranger* (she), enters a partnership with an established individual, *the insider* (he), whose type is public information, so that its reputation remains fixed. The stranger knows her type $\theta \in \{n,r\}$, and her reputation x summarizes the insider's posterior belief that she is rogue. The definitions of the collaboration regions, informational regions, and survival drift carry over to this one-dimensional environment, with x as the sole state variable.

In this one-sided environment, Proposition 1 describes the dynamics of the single state variable x. Equation (1) characterizes its law of motion from the stranger's perspective, while equation (2) captures the insider's view of the same process.

I focus on games where the insider is normative, since it is the only one of economic interest. If the insider were rogue, either type of stranger would have a profitable terminal

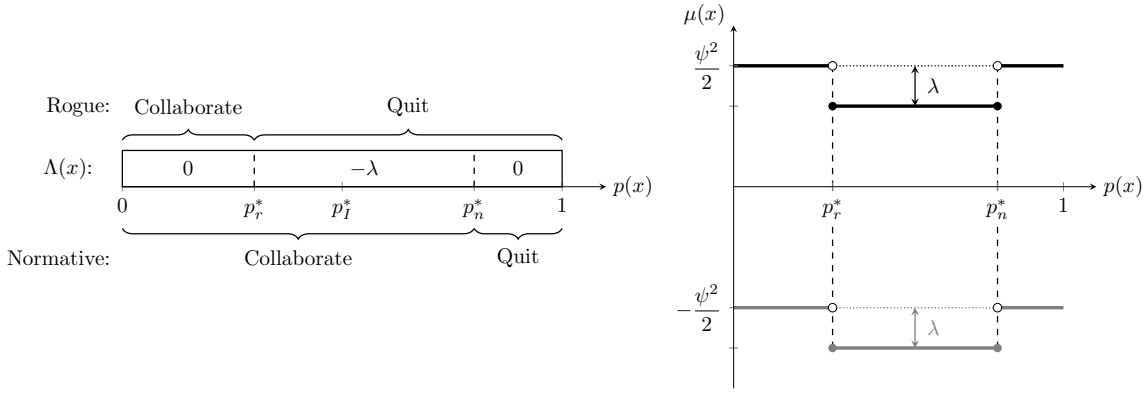


Figure 4: **One-Sided Equilibrium.** Left panel: braces indicate each stranger type's action at a stopping opportunity, and the strip reports the survival drift $\Lambda(x)$. All cutoffs are stated in belief space: $p_\theta^* \equiv p(x_\theta^*)$ for $\theta \in \{r, n\}$ and $p_I^* \equiv p(x_I^*)$. Right panel: reputation drift $\mu(x)$ of a rogue (black) and a normative (gray) stranger in equilibrium. Parameters: $\pi = 0.03$, $\rho = 0.1$, $\lambda = 0.1$, $\psi = 0.7$, $F_n = L_n = 0.73$, $L_r = 0.97$, $G_n = 1.37$.

action available, making the equilibrium trivial. With a normative insider, by contrast, the stranger's gains come entirely from the flow payoff π. A rogue stranger weighs this payoff against the risk that the signals reveal her type, while a normative stranger weighs it against the risk that they misrepresent it. In either case, the stranger has an incentive to collaborate until the risk of being called out becomes too large.

A strategy profile is a *cutoff profile* if there exist thresholds (x_n, x_r, x_I) such that, at each decision opportunity, a stranger of type θ collaborates when $x < x_\theta$ and quit otherwise, while the insider collaborates when $x < x_I$ and whistleblow otherwise. The following proposition characterizes equilibrium in terms of these three cutoffs.

Proposition 2. *The normative insider game has a unique cutoff Markov equilibrium, with cutoffs* (x_n^*, x_r^*, x_I^*) *satisfying* $x_r^* < x_n^*$.

Because $L_n \leq L_r$, the rogue stranger's cutoff always lies below the normative stranger's, $x_r^* < x_n^*$, so characterizing the equilibrium reduces to locating the insider's cutoff relative to this pair. Each location yields a different pattern of exit and accusation as the stranger's reputation deteriorates. The equilibrium exhibits *early accusation* if the insider's cutoff lies below both strangers' cutoffs, so that his threat starts before either type would quit; *sorting* if it lies between them, so that the rogue quits before the threat starts while the normative stays on despite it; and *early exit* if it lies above both, so that both types quit before the threat starts. Figure 4 illustrates the sorting equilibrium.

The primitives of the game determine which pattern arises. To see how, let $\bar{L}(\mu)$ denote the critical loss at which a stranger whose reputation has constant drift μ is

indifferent between quitting and collaborating at the insider's cutoff:

$$\bar{L}(\mu) = \frac{\pi}{\lambda}\left[1 + \frac{\sqrt{\mu^2 + 2\psi^2(\rho + 2\lambda)} - \mu}{\sqrt{\mu^2 + 2\psi^2\rho} + \mu}\right].$$

A stranger with loss L_θ and constant drift μ prefers to collaborate at the insider's cutoff if $L_\theta < \bar{L}(\mu)$, so its own cutoff lies above x_I^*, and prefers to quit there if $L_\theta > \bar{L}(\mu)$, so its cutoff lies below x_I^*. The threshold is decreasing in μ, since the numerator of the fraction falls and its denominator rises with μ. A stranger whose reputation drifts up faster spends more time in the insider's stop region, where the insider's call out arrives at rate λ, so a smaller loss suffices to make quitting optimal.

In equilibrium, however, the drift is not constant. A rogue stranger's reputation drifts at $\psi^2/2+\Lambda$ and a normative stranger's at $-\psi^2/2+\Lambda$, and the survival drift Λ varies with the state. Between the two strangers' cutoffs, only the rogue quits when an opportunity arrives, so continued collaboration is evidence that the stranger is normative and $\Lambda = -\lambda$; elsewhere, both types act alike and $\Lambda = 0$. Each stranger's drift therefore switches between two values, and its actual critical loss lies between the values of $\bar{L}$ at those two drifts. Applying these bounds to both types yields the equilibrium characterization,

Proposition 3. *The unique equilibrium exhibits early accusations if $L_\theta < \bar{L}(\psi^2/2)$, sorting if $L_n < \bar{L}(-\psi^2/2)$ and $\bar{L}(\psi^2/2 - \lambda) < L_r$, and early exit if $\bar{L}(-\psi^2/2 - \lambda) < L_\theta$.*

4.1. Partnership Dissolution

In any equilibrium, the game ends in finite time with positive probability. Let $\mathbb{P}_\theta^x$ denote the law of equilibrium play starting from an ongoing partnership with reputation x, conditional on the stranger being of type θ, and write $\mathbb{E}_\theta^x$ for the corresponding expectation. Let $(T_k^I)_{k\geq 1}$ and $(T_k^S)_{k\geq 1}$ denote the opportunity times of the insider and the stranger, respectively, and define the stopping times

$$\tau_I := \inf\{T_k^I : X_{T_k^I} > x_I^*\},$$
$$\tau_\theta := \inf\{T_k^S : X_{T_k^S} > x_\theta^*\},$$

with the convention $\inf\emptyset = \infty$. The partnership ends at $\tau = \tau_I \wedge \tau_\theta$. Since the two opportunity processes are independent, $\tau_I \neq \tau_\theta$ almost surely on $\{\tau < \infty\}$, so every

dissolution is either a call-out or a quit. Let

$$\omega_\theta(x) := \mathbb{P}^x_\theta(\tau_I < \tau_\theta),$$
$$q_\theta(x) := \mathbb{P}^x_\theta(\tau_\theta < \tau_I),$$

denote the probabilities that the insider ends the partnership by whistleblowing and that the stranger ends it by quitting, respectively. Their sum, $\omega_\theta(x) + q_\theta(x) = \mathbb{P}^x_\theta(\tau < \infty)$, is the probability that the partnership dissolves.

How much collaboration the equilibrium sustains depends on how long the partnership lasts. Expected duration, however, is uninformative: when the stranger is normative, her reputation drifts down, so the partnership survives forever with positive probability, $\omega_n(x) + q_n(x) < 1$, and $\mathbb{E}^x_n[\tau] = \infty$. Conditioning on dissolution instead would discard precisely the partnerships that last longest. Measuring collaboration by its discounted value avoids both problems. Define the *value of the partnership* as the expected discounted flow payoff that each partner accrues until termination,

$$v_\theta(x) := \mathbb{E}^x_\theta\left[\int_0^\tau e^{-\rho t}\pi\, dt\right] = \frac{\pi}{\rho}\left(1 - \mathbb{E}^x_\theta\left[e^{-\rho\tau}\right]\right),$$

where $e^{-\rho\tau} = 0$ on $\{\tau = \infty\}$. The value of the partnership is the flow payoff times the expected discounted duration. Unlike expected duration, it is always finite, and it increases as dissolution is delayed. An everlasting partnership has discounted duration $1/\rho$, so its value is π/ρ. At the other extreme, when both partners stop as soon as they can, the partnership ends at the first opportunity of either, which arrives at rate 2λ, so its value is $\pi/(\rho + 2\lambda)$. Hence

$$\frac{\pi}{\rho + 2\lambda} \leq v_\theta(x) \leq \frac{\pi}{\rho}.$$

From left to right, the panels of Figure 5 show how the stranger's initial reputation affects the probability of a call-out and the value of the partnership. A rogue stranger's reputation drifts up, and since no one ends the partnership below p^*_r, her reputation reaches p^*_r almost surely. Hence ω_r is constant below p^*_r. Above p^*_r, the rogue wants to quit but must wait for an opportunity, and before one arrives her reputation may cross the insider's cutoff, exposing her to being called out. The closer she starts to p^*_I, the more likely she loses this race, so ω_r rises. Above p^*_I, both partners stop as soon as they can, and their opportunities arrive at the same rate, which suggests $\omega_r = 1/2$. Yet ω_r stays below $1/2$, because the rogue's reputation can fall back into (p^*_r, p^*_I), where

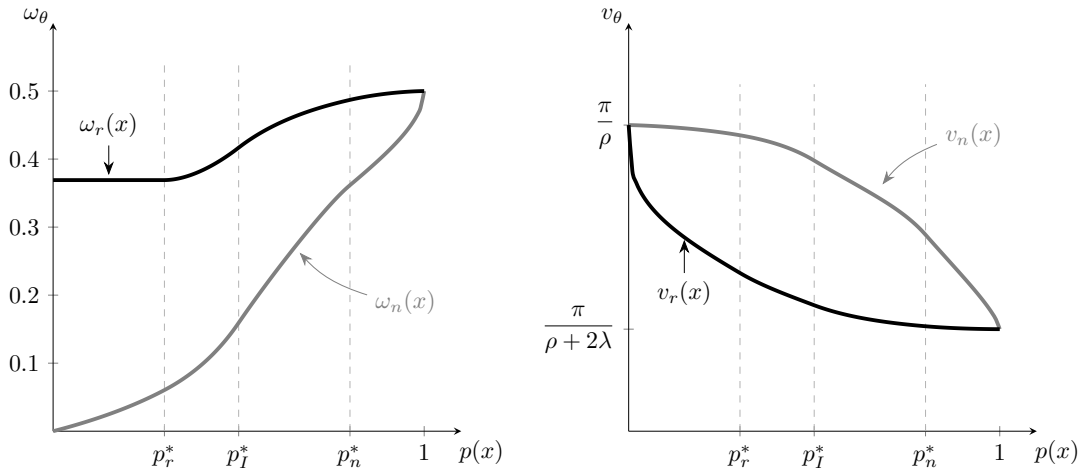


Figure 5: **Whistleblow and Partnership Value.** The left panel plots $\omega_\theta(x)$ and the right panel plots $v_\theta(x)$, both against the insider belief $p(x)$ for the equilibrium of Figure 4, with rogue strangers in black and normative strangers in gray.

only she stops. The higher the initial suspicion, the less likely such a return, so ω_r rises toward $1/2$ as $p(x) \to 1$. In contrast, a normative stranger's reputation drifts down, so suspicion that starts low is unlikely ever to reach the insider's cutoff, and ω_n vanishes as $p(x) \to 0$. However, $\omega_n > 0$ at every belief, and so false accusations occur with positive probability in equilibrium.

Suspicion erodes the collaboration that the partnership sustains. For a rogue stranger, dissolution occurs almost surely, so the value of the partnership depends only on how soon it ends. Because her reputation drifts toward the stop regions, the end comes sooner the higher she starts, and v_r falls quickly even at low beliefs. For a normative stranger, by contrast, dissolution is avoidable: her reputation drifts down, so misunderstandings tend to clear up before it reaches a stopping region, and v_n stays close to π/ρ at low beliefs. Only near a stop region, where a misunderstanding has little time to clear up, does v_n fall sharply.

Proposition 4. *In equilibrium:*

(i) Only partnerships with normative strangers can last forever.

(ii) The partner with the lower cutoff is more likely to end the partnership.

(iii) However low the initial suspicion, a normative stranger is falsely accused with positive probability.

(iv) The value of the partnership strictly decreases with suspicion and is higher with a normative stranger.

5. Equilibrium

5.1. Best Response and Optimal Stopping

The best response is characterized indirectly through the value function, from which the stopping regions are recovered. For a player of type θ facing a partner using strategy α, the value function

$$V_\theta(x, y; \alpha) := \sup_{\varphi} J_\theta(x, y; \varphi, \alpha). \tag{3}$$

A key observation is that the optimal terminal action depends only on the partner's current reputation y. The optimal stopping action therefore solves a static decision problem. Denote the *myopic action* and *myopic payoff* of type θ by

$$m_\theta(y) \in \operatorname*{argmax}_{a \in \mathcal{A}_\theta \setminus \{c\}} \overline{u}(a; y), \quad M_\theta(y) = \max_{a \in \mathcal{A}_\theta \setminus \{c\}} \overline{u}(a; y),$$

where

$$\overline{u}(a; y) = p(y)u(a; r) + (1 - p(y))u(a; n)$$

is the expected terminal payoff given the partner's reputation y. Because each type has only two terminal actions available, the myopic action admits a threshold characterization:

$$m_n(y) = \begin{cases} w & y \geq y_n \\ q & y < y_n \end{cases}; \qquad m_r(y) = \begin{cases} s & y \geq y_r \\ q & y < y_r \end{cases}; \qquad \begin{aligned} y_n &= \log(F_n/G_n) \\ y_r &= \log(L_r/G_r) \end{aligned}$$

A strategy α is called *myopically optimal* if, for every type θ, its decision rule prescribes the myopic action whenever the player stops:

$$\alpha_\theta(x, y) = m_\theta(y) \quad \text{for all } (x, y) \notin \mathcal{C}_\theta(\alpha).$$

Proposition 5. *For any strategy α, there exists a unique bounded continuous value function $V_\theta(\cdot; \alpha)$, attained by a myopically optimal decision rule φ^*_θ with collaboration region*

$$\mathcal{C}^*_\theta = \{(x, y) : V_\theta(x, y; \alpha) > M_\theta(y)\}.$$

Players continue collaborating, gathering information and receiving the flow payoff, until they stop, at which point they select their terminal action myopically. Collaboration, however, is risky: normative players may be mistaken for rogues, while rogue players may be exposed. At each stopping opportunity, a player compares the forward-

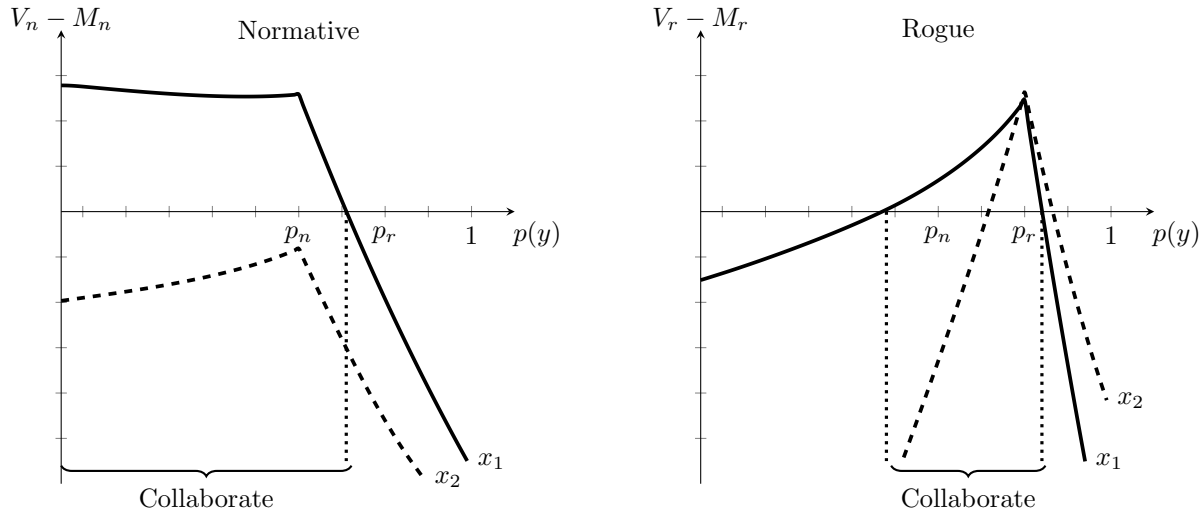


Figure 6: **Collaboration Decision**. For each type $\theta \in \{n, r\}$, the figure plots $V_\theta(x, y) - M_\theta(y)$ at two fixed levels of own reputation, $x_1 < x_2$ (solid and dashed, respectively). The positive portion of each curve identifies the cross-section of the collaboration region. The thresholds $p_n \equiv p(y_n)$ and $p_r \equiv p(y_r)$ denote the myopic stopping thresholds.

looking value of collaboration, V_θ, with the myopic payoff from stopping, M_θ. Figure 6 illustrates this comparison by plotting their difference, $V_\theta(x, y) - M_\theta(y)$, at two levels of own reputation, $x_1 < x_2$.

At the lower own-reputation state x_1, normative players collaborate unless the partner's reputation is sufficiently high, in which case whistleblowing becomes optimal. Rogue players, by contrast, collaborate only when their partner's reputation lies in an intermediate range. When it is low, the expected evolution of reputations depresses their continuation value: by Proposition 1, a rogue player anticipates their own reputation drifting upward and y downward. Because the partner may receive the next opportunity first, this unfavorable drift lowers the value of waiting, and quitting is optimal. Higher values of y make a normative type less likely, weakening this concern until collaboration is optimal. Once y is sufficiently high, the rogue player switches to scheming.

At the high own-reputation state x_2, the prospect that the partner ends the relationship matters more. For normative players, this unambiguously lowers the collaboration value. The signal component of their reputation drift works in their favor, but the survival drift works against them at high reputations, where only rogue types collaborate. A normative player's reputation is therefore unlikely to improve before the partner receives an opportunity to call her out or scheme. For rogue players, the effect depends on y. At low y, where the odds favor a normative type, moving from x_1 to x_2 makes calling out more likely: the collaboration value falls, and the collaboration region contracts from

the left. At high y, where the odds favor a rogue type, the same move makes scheming more likely: the collaboration value rises, and the region expands to the right.

Figure 6 also showcases how the present problem departs from standard continuous-time optimal stopping. There, termination is available at every instant, so optimality at the boundary of the stopping region requires local indifference between stopping immediately and delaying by an arbitrarily short interval, which yields the smooth-fit condition; otherwise, the agent would strictly prefer to stop slightly earlier or slightly later. This would lead $V_\theta - M_\theta$ to cross zero tangentially. In contrast, here once an opportunity is declined the player cannot reconsider an instant later and must wait for the next opportunity. This eliminates the local timing deviation underlying smooth fit, and the crossing occurs transversely.

5.2. Equilibrium Characterization

Given any conjecture about the partner's stopping behavior, a player's optimal response is unique. The reputation dynamics governing that response, however, depend on the conjectured behavior itself through the survival drift. An equilibrium must therefore be self-confirming: stopping behavior must be optimal under the reputational consequences that it generates. The following proposition establishes that, when stopping opportunities are sufficiently infrequent, this consistency requirement admits a unique solution.

Proposition 6. *There exists $\bar{\lambda} > 0$ such that, for every $\lambda \in (0, \bar{\lambda})$, the game admits a unique symmetric Markov perfect equilibrium.*

The restriction on λ limits two channels through which strategic feedback may generate multiple self-consistent stopping patterns. First, frequent opportunities strengthen preemption: when players can react almost immediately to changes in reputation, a player approaching a region in which her partner would act against her may quit before reaching it. This anticipated stopping region thus shapes exit before players reach it, and earlier exit changes what survival reveals.[6] Second, frequent opportunities make survival a stronger signal of type, due to the survival drift being proportional to λ. Thus differences in stopping behavior can become the main force shaping reputation evolution when λ is large relative to $\psi^2/2$.

Two notions describe the shape of the collaboration regions. Type θ *exhibits fear* on an interval (x_1, x_2), possibly unbounded, if she ends the game at the first opportunity

[6]Continuous-time timing games may lack pure-strategy equilibria or admit multiple equilibria; see Fudenberg and Tirole (1985); Riedel and Steg (2017) for preemption and Hendricks et al. (1988); Myatt (2025) for attrition games.

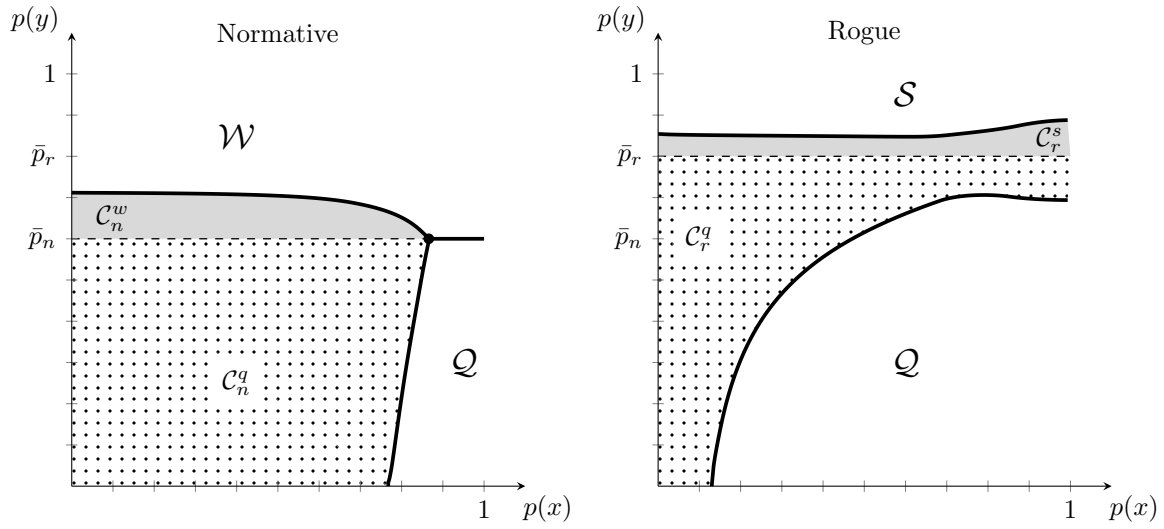


Figure 7: **Equilibrium.** Shaded areas denote collaboration regions, partitioned according to the myopically optimal terminal action that replaces: $\mathcal{C}_\theta^a = \{(x,y) \in \mathcal{C}_\theta : m_\theta(y) = a\}$. Parameters: $\pi = 0.03$, $\rho = 0.1$, $\lambda = 0.1$, $\psi = 0.7$, $F_n = L_n = 3/2$, $L_r = 4$, $G_n = G_r = 1$.

at which her own reputation lies in that interval, whatever her partner's reputation is. Type θ *exhibits openness* on (x_1, x_2) if, for some partner reputation y, she collaborates at every own reputation in that interval,

$$\begin{aligned} \textit{Fear} \text{ on } (x_1,x_2): &\quad \mathcal{C}_\theta \cap \big((x_1,x_2)\times\mathbb{R}\big) = \varnothing, \\ \textit{Openness} \text{ on } (x_1,x_2): &\quad (x_1,x_2)\times\{y\} \subseteq \mathcal{C}_\theta \text{ for some } y \in \mathbb{R}. \end{aligned}$$

Fear captures terminations that are triggered by her own reputation, within the interval, no partner reputation is reassuring enough to sustain collaboration. Openness, by contrast, means that with the right partner, no reputation in the interval prompts her to end the game. The two notions are mutually exclusive: a type that exhibits fear on an interval cannot exhibit openness on any interval that overlaps it.

Proposition 7. *There exist a threshold $\bar{L}_n^\infty > 0$ such that the unique equilibrium has the following property:*

(i) Normative players exhibit fear on $(\bar{x}, \infty)$ for some finite $\bar{x}$ if $L_n > \bar{L}_n^\infty$, and exhibit openness on $\mathbb{R}$ if $L_n < \bar{L}_n^\infty$.

Figure 7 plots an example of the unique equilibrium when the normative type exhibit fear. The next proposition characterize the distribution of types in the surviving partnerships

Proposition 8. *In the unique equilibrium, mixed partnerships dissolve almost surely, whereas same-type partnerships last forever with positive probability.*

6. Discussion

In this model, reputational concerns destroy partnerships rather than information. These concerns arise because the organization in which partnerships develop enforces a social norm, and players differ in whether they adhere to it. Yet adherence to the norm is irrelevant to joint output. Partnerships therefore end for reasons that have nothing to do with their productivity. The model suggests that this enforcement not only purges those who defy the norm but also generates false accusations in equilibrium. A lower cost of false accusations makes normative players act on weaker evidence and their partners more fearful, lowering the value of every partnership. Beyond these direct effects, partnership survival sorts the organization's members, so small cliques of rogues emerge.

Three simplifications qualify this model. First, talk is exogenous: the signals capture what players reveal unintentionally, not what they choose to say, so the model shuts down the channel by which reputational concerns destroy information (Morris, 2001; Ely and Välimäki, 2003). Second, the uniqueness result for the two-sided game covers only rare opportunities. When opportunities are frequent, preemption and learning from survival may sustain several equilibria, which might help explain how abruptly reputation concerns can escalate into a witch hunts. Finally, the analysis follows a single pair. In an organization, workers whose collaboration ends would seek new matches, and assortative survival suggests that reputation concern would gradually segregate the organization by type.

A. Technical Foundations

All stochastic processes are defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The main text treats the reputation profile (X^i, X^j) as a well-posed Markov diffusion and each value function V_θ as an almost-everywhere differentiable solution of a dynamic-programming equation. This appendix establishes both. The starting point is the well-posedness of the system introduced in Proposition 1.

Lemma A.1. *The system of SDEs in Proposition 1 admits a unique strong solution. Moreover, the resulting process is strong Markov and strong Feller.*

The survival drift in Proposition 1 is bounded—$|\Lambda| \leq \lambda$ and the signal terms are bounded—but discontinuous, since Λ jumps across the boundaries of the collaboration regions. Well-posedness rests on the additive, nondegenerate noise: with constant diffusion coefficient ψ and bounded measurable drift, the system admits a unique strong solution by Veretennikov (1981). In addition, nondegeneracy ensures that the transition semigroup admits a density, so the solution is strong Feller, and time-homogeneity gives the strong Markov property. These properties support the rest of the analysis: the strong Markov property licenses the dynamic-programming representation in Appendix A.1; the strong Feller property guarantees that V_θ inherits continuity. In addition, the existence of a density implies the process spends zero time on any null set, which is what renders the discontinuities of Λ immaterial for the dynamics.

A.1. HJB Equation and Regularity

Proposition 5 characterizes $V_\theta(\cdot; \alpha)$ as the value of an optimal stopping problem in which stopping is restricted to the player's Poisson opportunities. Its dynamic-programming counterpart is recorded next; this is the representation used to recover the stopping geometry and to differentiate the value function. Given the strategy α and a bounded measurable candidate continuation value f, define the operator capturing the value at the partner's opportunity:

$$\begin{aligned}\Gamma_\theta(x, y; f) := p(y) \left[\mathbb{1}\{(y, x) \in \mathcal{C}_r(\alpha)\} f(x, y) + \mathbb{1}\{(y, x) \notin \mathcal{C}_r(\alpha)\} u^p(\alpha_r(y, x); \theta)\right] \\ + (1 - p(y)) \left[\mathbb{1}\{(y, x) \in \mathcal{C}_n(\alpha)\} f(x, y) + \mathbb{1}\{(y, x) \notin \mathcal{C}_n(\alpha)\} u^p(\alpha_n(y, x); \theta)\right].\end{aligned}$$

The first bracket conditions on a rogue partner. If she collaborates at state (y, x), player i's value remains $f(x, y)$; otherwise, she ends the game at her first opportunity with action $\alpha_r(y, x)$, and player i receives $u^p(\alpha_r(y, x); \theta)$. The second bracket does the same for a normative partner.

The value function $V_\theta(\cdot;\alpha)$ from Proposition 5 is characterized as the solution of the Hamilton–Jacobi–Bellman equation

$$\rho V_\theta = \pi + \mathcal{L}_\theta V_\theta + \lambda\big(M_\theta(y) - V_\theta\big)^+ + \lambda\big(\Gamma_\theta(x,y;V_\theta) - V_\theta\big). \tag{A.1}$$

Equation (A.1) admits a natural asset-pricing interpretation. The value V_θ can be viewed as the value of an asset that pays the dividend π while the partnership remains active through collaboration. Changes in the reputation profile generate the expected capital gain $\mathcal{L}_\theta V_\theta$. In addition, liquidation opportunities arrive at rate λ. At the player's own opportunity, the asset is liquidated for M_θ whenever liquidation is optimal, producing the term $\lambda(M_\theta - V_\theta)^+$. Similarly, at the partner's opportunity, control over liquidation passes to them, with the resulting value summarized by $\lambda(\Gamma_\theta - V_\theta)$. Thus, the discounted value ρV_θ equals the dividend plus expected capital gains from reputation dynamics and from the possibility that either player ends the partnership.

Because stopping is constrained to the opportunity process, a player cannot exercise the stopping option immediately at an arbitrary state. As a result, M_θ does not impose the pointwise constraint $V_\theta \geq M_\theta$ familiar from standard continuous-time optimal stopping, and the usual smooth-pasting condition need not hold at the stopping boundary. In addition, the indicator functions entering Γ_θ generate discontinuities across the partner's stopping boundaries, so classical C^2 solutions need not exist. This motivates the next result.

Lemma A.2 (Regularity)**.** *Let α be a myopically optimal strategy and fix $\theta \in \{n, r\}$. The value function $V_\theta(\cdot;\alpha)$ defined in* (3) *satisfies:*

(i) $V_\theta(\cdot;\alpha) \in W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2)$ for every $p \in (1,\infty)$;

(ii) $V_\theta(\cdot;\alpha) \in C^{1,\gamma}_{\mathrm{loc}}(\mathbb{R}^2)$ for every $\gamma \in (0,1)$;

(iii) for every $p > 2$, $V_\theta(\cdot;\alpha)$ is the unique bounded $W^{2,p}_{\mathrm{loc}}$ strong solution of (A.1).

The right-hand side of (A.1) is bounded and hence belongs to $L^p_{\mathrm{loc}}(\mathbb{R}^2)$ for every $p < \infty$. Since $\mathcal{L}_\theta$ is uniformly elliptic, standard interior L^p estimates imply that $V_\theta \in W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2)$, establishing (i). In two dimensions, $W^{2,p}_{\mathrm{loc}}$ embeds into $C^{1,\gamma}_{\mathrm{loc}}$ whenever $\gamma < 1 - 2/p$. Because (i) holds for arbitrarily large p, it follows that $V_\theta \in C^{1,\gamma}_{\mathrm{loc}}(\mathbb{R}^2)$ for every $\gamma \in (0,1)$, proving (ii). The substantive gain over Proposition 5 is therefore the upgrade from continuity to continuous differentiability, despite the discontinuities generated by the stopping rules in Γ_θ. Finally, (iii) identifies the probabilistically defined value function (3) with the strong solution of the HJB equation (A.1), so the probabilistic and analytic formulations of the best-response problem coincide.

Lemma A.3 (Transversality)**.** *Let α be a myopically optimal strategy. For each $\theta \in \{n, r\}$, the level set $\{(x, y) \in \mathbb{R}^2 : V_\theta(x, y; \alpha) = M_\theta(y)\}$ has measure zero. In particular $\partial \mathcal{C}_\theta(\alpha)$ has measure zero.*

The nullity of the indifference set rules out thick regions in which players are indifferent between collaborating and stopping. This makes behavior on the boundary immaterial for the dynamics: although Λ and Γ_θ depend on indicators of the collaboration regions, their values on the boundary affect only a null set of states. Together with the transition density from Lemma A.1, this nullity implies that the reputation process spends zero time on the indifference set.

Lemma A.4 (Stability)**.** *Consider two possible strategies α and $\widetilde{\alpha}$ employed by the partner. Suppose $\|V_\theta(\cdot; \alpha) - V_\theta(\cdot; \widetilde{\alpha})\|_\infty \leq \varepsilon$ for each $\theta \in \{n, r\}$. Then the optimal decision rules $\varphi^*_\theta(\cdot; \alpha)$ and $\varphi^*_\theta(\cdot; \widetilde{\alpha})$ can differ only on the ε-indifference region:*

$$\{(x, y) \in \mathbb{R}^2 : \varphi^*_\theta(x, y; \alpha) \neq \varphi^*_\theta(x, y; \widetilde{\alpha})\} \subseteq \{(x, y) : |V_\theta(x, y; \alpha) - M_\theta(y)| \leq \varepsilon\}.$$

Equivalently,

$$\{(x, y) \in \mathbb{R}^2 : \varphi^*_\theta(x, y; \alpha) = \varphi^*_\theta(x, y; \widetilde{\alpha})\} \supseteq \{(x, y) : |V_\theta(x, y; \widetilde{\alpha}) - M_\theta(y)| > \varepsilon\}$$

Away from the indifference, the optimal action is robust to small changes in the value function. If $V_\theta(x, y; \alpha)$ lies more than ε above or below M_θ, then any $V_\theta(x, y; \widetilde{\alpha})$ within ε has the same sign relative to M_θ, so the optimal decision rule φ^*_θ prescribes the same action. A change in behavior is therefore possible only near the indifference set. Lemma A.3 ensures that these near-indifference regions become negligible as ε shrinks. Hence, uniform convergence of value functions carries over to stability of the induced strategies.

B. Solution of the normative insider game

This appendix derives the results of a stranger facing a normative insider.

B.1. Equilibrium

Let $\alpha = (\alpha_n, \alpha_r)$ describe the stranger's conjectured decision rules and β the insider's. Since the insider is known to be normative, the stranger's decision rule is restricted to collaborate or quit, as scheme and whistleblow are strictly dominated. Similarly, the insider's decision rule is restricted to collaborate or whistleblow, since quit is dominated

by collaboration. The reputation process follows from Proposition 1, which specializes to

$$dX_t = \left(\Lambda(X_t;\alpha) + \frac{\psi^2}{2}\big(\mathbb{1}\{\theta = r\} - \mathbb{1}\{\theta = n\}\big)\right) dt + \psi dB_t^i, \tag{B.2}$$

$$dX_t = \left(\Lambda(X_t;\alpha) + \frac{\psi^2}{2}\tanh\left(\frac{X_t}{2}\right)\right) dt + \psi\, d\widetilde{B}_t^j, \tag{B.3}$$

where $\theta \in \{n, r\}$ is the stranger's type. Equation (B.2) conditions on the stranger's information set, while equation (B.3) uses the insider's information set. For each type $\theta \in \{n, r\}$, the stranger's Bellman equation is

$$\rho V_\theta = \pi + \frac{\psi^2}{2}V_\theta'' + \left(\Lambda + \frac{\psi^2}{2}\big(\mathbb{1}\{\theta = r\} - \mathbb{1}\{\theta = n\}\big)\right) V_\theta' + \lambda(-V_\theta)^+ - \lambda(V_\theta + L_\theta)\mathbb{1}\{\beta = w\}, \tag{B.4}$$

while the insider's Bellman equation is

$$\rho W = \pi + \frac{\psi^2}{2}W'' + \left[\Lambda + \frac{\psi^2}{2}\tanh(x/2)\right] W' + \lambda[(M_n - W)^+ - HW],$$

where $H(x) = p(x)\mathbb{1}\{\alpha_r = q\} + (1 - p(x))\mathbb{1}\{\alpha_n = q\} \in [0, 1]$ is the probability that the stranger ends the game given an opportunity, and $M_n(x) = p(x)[G_n + F_n] - F_n$ is the expected payoff from insider whistleblowing.

Fix a finite insider cutoff x_I and conjectured stranger cutoff decision rules α. The conjectured cutoffs determine Λ, while V_θ is the optimized value against that drift and insider cutoff. Its zero need not coincide with the type's conjectured cutoff until consistency is imposed. To characterize the stranger's optimal response, the first step is to show that her continuation value $V_\theta(x)$ is non-increasing in reputation x. Since quitting yields zero, this implies that the region in which collaboration is strictly optimal is a lower interval.

Construct successive continuation values, starting from zero. Each update computes the expected discounted flow until the next opportunity and the discounted value at that opportunity. At the stranger's own opportunity, the value is the maximum of the preceding continuation value and zero. At an insider opportunity, the value to the stranger is continuation below x_I and $-L_n$ if normative, or $-L_r$ if rogue, above it. If the preceding continuation value is non-increasing and bounded below by the corresponding terminal payoff, both functions are non-increasing: the switch at x_I cannot create an upward jump. The lower bound is preserved by every update because the flow payoff is positive and terminal payoffs are bounded below by $-L_\theta$.

Holding the conjectured strategy fixed, the one-dimensional reputation dynamics preserve the ordering of initial reputations when paths are compared under the same shocks. Consequently, taking discounted expectations of the non-increasing opportunity payoffs preserves monotonicity in the initial reputation x. The discounted Bellman operator resulting from this process is a contraction, as shown later in the proof of Proposition 5. Its iterates therefore converge to V_θ, which inherits their monotonicity and is thus non-increasing in x. The tail limits of V_θ are

$$\lim_{x\to-\infty} V_\theta(x) = \frac{\pi}{\rho} > 0, \qquad \lim_{x\to+\infty} V_\theta(x) = \frac{\pi - \lambda L_\theta}{\rho + 2\lambda}. \tag{B.5}$$

Hence, continuity implies that V_θ has a zero. Since V_θ is non-increasing, its zero set is an interval. Furthermore, that interval cannot have a positive length, as the Bellman equation (B.4) would reduce, on a subinterval not containing x_I, to

$$0 = \pi - \lambda L_\theta \mathbb{1}\{x > x_I\}.$$

This identity cannot hold: below x_I it requires $\pi = 0$, and above x_I it requires $\pi = \lambda L_\theta$, which contradicts the assumption $\pi < \lambda L_\theta$. Thus, each type satisfying its condition has a unique zero, that is, there is a unique reputation x_θ such that $V_\theta(x_\theta) = 0$. Thus, a stranger of type θ optimally chooses to collaborate for all reputations $x < x_\theta$ and quits at all reputations $x \geq x_\theta$, given the opportunity. Since the normative stranger has a lower reputation drift and a smaller loss, $L_n < L_r$, $V_n(x) > V_r(x)$, which implies $x_n > x_r$.

A equilibrium must satisfy the following Bellman equations and matching conditions: the strangers' Bellman equations

$$\begin{aligned}
\rho V_n &= \pi + \frac{\psi^2}{2} V_n'' + \left(\Lambda - \frac{\psi^2}{2}\right) V_n' - \lambda \left[\mathbb{1}\{x > x_n^*\} V_n + \mathbb{1}\{x > x_I^*\}(V_n + L_n)\right], \\
\rho V_r &= \pi + \frac{\psi^2}{2} V_r'' + \left(\Lambda + \frac{\psi^2}{2}\right) V_r' - \lambda \left[\mathbb{1}\{x > x_r^*\} V_r + \mathbb{1}\{x > x_I^*\}(V_r + L_r)\right],
\end{aligned}$$

the insider's Bellman equation

$$\rho W = \pi + \frac{\psi^2}{2} W'' + \left[\Lambda + \frac{\psi^2}{2}\tanh(x/2)\right] W' + \lambda[\mathbb{1}\{x > x_I^*\}(M_n - W) - HW],$$

and the survival drift

$$\Lambda(x; \alpha^*) = \lambda[\mathbb{1}\{x > x_n^*\} - \mathbb{1}\{x > x_r^*\}],$$

with value matching conditions $V_\theta(x_\theta^*) = 0$ and $W(x_I^*) = M_n(x_I^*)$, and the values and first derivatives are continuous across the cutoffs.

Next, I show that, for each fixed finite insider cutoff x_I, there is exactly one consistent stranger pair. Fix a finite insider cutoff x_I. Since $|\Lambda| \leq \lambda$, the reputation drift is uniformly bounded over conjectured cutoffs. Over any fixed time interval, the probability of crossing x_I vanishes uniformly as the initial reputation approaches either extreme. Bounded payoffs and discounting therefore make the tail limits in (B.5) uniform over conjectures. Moving a conjectured cutoff changes the drift only on a shrinking interval. The corresponding discounted occupation tends to zero for this nondegenerate diffusion, so the optimized values depend continuously on the conjectured cutoffs.

For each conjectured x_n, there is at most one consistent rogue cutoff. At a zero of the rogue consistency condition, the rogue's value under that conjecture is strictly negative at every higher reputation. A higher conjectured rogue cutoff raises Λ, weakly lowering her value at each fixed reputation. Its consistency value is therefore strictly negative, ruling out two consistent rogue cutoffs. Continuity and the positive and negative tail limits guarantee existence. Denote the resulting cutoff by $x_r(x_n)$. Uniqueness, continuity of the consistency condition, and the uniform tail limits imply that this response is continuous.

Along this consistent rogue response, the normative value evaluated at the conjectured x_n is continuous. The same uniform tail limits make it positive for sufficiently low conjectured cutoffs and negative for sufficiently high conjectured cutoffs. It therefore has at least one zero. Consider any such zero, so both strangers are consistent. The regularity argument in Appendix A gives continuously differentiable values, smooth between the cutoffs. Differentiating their Bellman equations with respect to reputation, away from the cutoffs, gives

$$\frac{\psi^2}{2}V_r''' + \left(\Lambda + \frac{\psi^2}{2}\right)V_r'' - [\rho + \lambda(\mathbb{1}\{x > x_r\} + \mathbb{1}\{x > x_I\})]V_r' = 0,$$

and

$$\frac{\psi^2}{2}V_n''' + \left(\Lambda - \frac{\psi^2}{2}\right)V_n'' - [\rho + \lambda(\mathbb{1}\{x > x_n\} + \mathbb{1}\{x > x_I\})]V_n' = 0,$$

Now substitute $e^{-x}V_n'$ into the differential equation satisfied by V_r', using

$$\Lambda = \lambda[\mathbb{1}\{x > x_n\} - \mathbb{1}\{x > x_r\}]$$

at the consistent pair,

$$\frac{\psi^2}{2}\left(e^{-x}V_n'\right)'' + \left(\Lambda + \frac{\psi^2}{2}\right)(e^{-x}V_n')' - [\rho + \lambda(\mathbb{1}\{x > x_r\} + \mathbb{1}\{x > x_I\})]e^{-x}V_n' = 0,$$

$$e^{-x}\left\{\frac{\psi^2}{2}V_n''' + \left(\Lambda - \frac{\psi^2}{2}\right)V_n'' - [\rho + \lambda(\mathbb{1}\{x > x_n\} + \mathbb{1}\{x > x_I\})]V_n'\right\} = 0,$$

Thus, V_r' and $e^{-x}V_n'$ satisfy the same second-order differential equation away from the cutoffs. Below all cutoffs, the common differential equation has the following characteristic equation and roots,

$$\xi^2 + \xi - \frac{2\rho}{\psi^2} = 0, \qquad \xi = \frac{-1 \pm \sqrt{1 + 8\rho/\psi^2}}{2}.$$

Boundedness of the original value function as $x \to -\infty$ excludes the negative root. Consequently, both V_r' and $e^{-x}V_n'$ are constant multiples of

$$\exp\left[\frac{-1 + \sqrt{1 + 8\rho/\psi^2}}{2}x\right].$$

To extend this proportionality across the stranger cutoffs, suppose first that the cutoffs are distinct. Subtract the left-hand limit of each stranger's Bellman equation from its right-hand limit. The values and first derivatives are continuous. Moreover, the change in a stranger's own quitting term contributes zero at her cutoff because $V_\theta(x_\theta) = 0$. Since Λ decreases by λ at x_r, and increases by λ at x_n,

$$\frac{\psi^2}{2}\left(\lim_{x\downarrow x_r} V_\theta''(x) - \lim_{x\uparrow x_r} V_\theta''(x)\right) - \lambda V_\theta'(x_r) = 0.$$
$$\frac{\psi^2}{2}\left(\lim_{x\downarrow x_n} V_\theta''(x) - \lim_{x\uparrow x_n} V_\theta''(x)\right) + \lambda V_\theta'(x_n) = 0.$$

Because $(e^{-x}V_n')' = e^{-x}(V_n'' - V_n')$, and V_n' is continuous, the jump in this derivative is e^{-x} times the jump in V_n''. Thus, the two functions obey the same matching rules: each is continuous, and its derivative jump at each stranger cutoff is proportional to its own value. If their values and derivatives are proportional immediately before a stranger cutoff, they remain proportional immediately after it, with the same factor. Uniqueness for their common second-order differential equation preserves proportionality throughout the next interval. Repeating this argument establishes proportionality throughout $x < x_I$.

On the right side, start from the right tail $x \to +\infty$. Above all cutoffs, the common

differential equation has the following characteristic equation and roots,

$$\xi^2 + \xi - \frac{2\rho + 4\lambda}{\psi^2} = 0, \qquad \xi = \frac{-1 \pm \sqrt{1 + 8(\rho + 2\lambda)/\psi^2}}{2}.$$

Boundedness selects the common decaying root. Then, applying the same argument backward through the stranger cutoffs establishes proportionality through $x > x_I$.

Finally, both functions are continuous at x_I and are strictly negative there, as follows. Monotonicity already gives $V'_\theta \leq 0$, so suppose $V'_\theta(x_I) = 0$. Then, V'_θ attains its maximum at x_I, so V''_θ can only jump downward here. But the Bellman equation gives

$$\lim_{x \downarrow x_I} V''_\theta(x) - \lim_{x \uparrow x_I} V''_\theta(x) = \frac{2\lambda}{\psi^2}[V_\theta(x_I) + L_\theta] > 0,$$

where positivity follows from

$$V_\theta(x) \geq \frac{\pi - \lambda L_\theta}{\rho + \lambda} > -L_\theta.$$

Indeed, always collaborating does at least as well as facing an insider who whistleblows at every opportunity; the payoff in that latter case is the displayed lower bound. The derivative cannot vanish at another finite reputation either. A zero of the non-positive function V'_θ is a maximum. Away from the cutoffs, its derivative must therefore be zero. At a stranger cutoff, the homogeneous jump condition also makes the two one-sided derivatives agree, so both must be zero. Uniqueness for the homogeneous differential equation and its matching conditions would then propagate this zero to x_I, contradicting the argument above. Hence $V'_\theta(x) < 0$ at every finite reputation. Therefore continuity at x_I makes the proportionality factors on its two sides coincide, yielding

$$\frac{V'_n(x)}{V'_n(x_I)} = e^{x - x_I} \frac{V'_r(x)}{V'_r(x_I)}. \tag{B.6}$$

If cutoffs coincide, their jump contributions are added. Proportionality is still propagated separately up to x_I from each tail.

The earlier comparison between the stranger types can now be strengthened. The normative stranger has a lower reputation drift, and $V'_r < 0$ makes the gain from that lower drift strictly positive: substituting V_r into the normative Bellman equation contributes $-\psi^2 V'_r(x) > 0$. The weakly smaller whistleblowing loss contributes the additional nonnegative term $\lambda \mathbb{1}\{x > x_I\}(L_r - L_n)$. Thus V_r is a strict subsolution of the normative Bellman equation, and strong comparison gives $V_n(x) > V_r(x)$. At x_r,

therefore, $V_n(x_r) > V_r(x_r) = 0$, so the unique zeros satisfy $x_r < x_n$.

Before differentiating with respect to conjectured cutoffs, local smooth dependence also needs justification. For prescribed conjectured and own quitting cutoffs, the values solve linear equations with continuous value and first-derivative matching. Their exponential solutions and matching conditions depend smoothly on the cutoff positions within each strict ordering. The matching system is nonsingular because the bounded discounted value for a prescribed policy is unique. At an optimal own quitting cutoff, moving that cutoff produces no first-order source because its value is zero. Since $V'_\theta(x_\theta) < 0$, the implicit function theorem gives local smooth dependence of the optimal cutoff and value. At rogue consistency, the derivative of its indifference condition with respect to the conjectured rogue cutoff is

$$V'_r(x_r) + \left.\frac{\partial V_r(x;\alpha)}{\partial x_r}\right|_{x=x_r} < 0 :$$

the first term is strictly negative, and the second is nonpositive because raising the conjectured rogue cutoff raises the drift. Hence $x_r(x_n)$ is locally differentiable at the consistent pair. When x_I coincides with a stranger cutoff, the same argument uses one-sided variations, with the coincident jump contributions added.

Equipped with the proportionality relation (B.6), I now bound the first-order change in the rogue stranger's value induced by increasing the conjectured normative cutoff x_n, while readjusting $x_r(x_n)$ to preserve the rogue's indifference at her cutoff. Throughout this comparison, the current reputation x and insider cutoff x_I remain fixed.

To bound the rogue's response to the changing cutoffs, I compare it with $-V'_r(x)$. At each reputation, consider the difference between $-V'_r(x)$ and the change in the rogue's value per unit increase in x_n, with $x_r(x_n)$ preserving the indifference,

$$-V'_r(x) - \frac{d}{dx_n}V_r(x;\alpha) = -V'_r(x) - \left[\frac{\partial V_r}{\partial x_n} + \frac{\partial V_r}{\partial x_r}\frac{dx_r(x_n)}{dx_n}\right]. \tag{B.7}$$

Between cutoffs, the indicators defining Λ are locally unchanged when the cutoff positions move. Differentiating the rogue's Bellman equation with respect to x_n therefore gives the same homogeneous equation satisfied by V'_r. By linearity, the difference (B.7) also satisfies this equation.

Suppose first that the cutoffs are distinct. Differentiating rogue indifference gives,

$$\left.\frac{dV_r(x;\alpha)}{dx_n}\right|_{x=x_r} + V'_r(x_r)\frac{dx_r}{dx_n} = 0.$$

Combining this equality with the differentiated continuity conditions shows that the difference is continuous and that its reputation derivative jumps at x_r by

$$\frac{2\lambda}{\psi^2}\left[-V_r'(x_r) - \left.\frac{dV_r(x;\alpha)}{dx_n}\right|_{x=x_r}\right]$$

At x_n, its derivative is continuous: the jump in the derivative of $-V_r'$ cancels the jump induced by moving x_n. At the fixed x_I, its derivative jumps by

$$-\frac{2\lambda}{\psi^2}\left[V_r(x_I) + L_r\right] < 0.$$

The difference also approaches zero at both extremes ($x \to +\infty$ and $x \to -\infty$). On each tail, V_r equals its corresponding limiting value in (B.5) plus a multiple of a decaying exponential. Changing the cutoffs changes the multiplier, while the exponent remains unchanged. Thus both V_r' and dV_r/dx_n approach zero.

These conditions imply that the difference is strictly positive. If it were negative somewhere, its zero tail limits would imply a negative minimum. Between cutoffs, such a minimum contradicts the homogeneous equation because the discount coefficient is positive. At x_r, a negative minimum would require a nonnegative derivative jump, whereas the prescribed jump is negative. The strictly negative jump at x_I likewise excludes a minimum there. At x_n, the derivative is continuous, so the same one-sided differential-equation argument applies.

A zero minimum is also impossible: uniqueness for the homogeneous equation would propagate zero value and zero derivative to x_I, contradicting the strictly negative jump there. Consequently, the difference in (B.7) is strictly positive at every finite reputation. If x_I coincides with a stranger cutoff, the jump contributions are added: at a negative or zero minimum, their combined jump has the same contradictory sign. One-sided variations therefore give the same conclusion.

Now consider any zero of the normative consistency condition, so that both stranger cutoffs are consistent. Apply the exponential substitution to the linearized Bellman equations for the value changes. Their moving-cutoff derivative jumps involve

$$-\frac{2\lambda}{\psi^2}V_\theta'(x_r)\frac{dx_r}{dx_n} \quad \text{and} \quad \frac{2\lambda}{\psi^2}V_\theta'(x_n).$$

Equation (B.6) makes these jumps proportional after multiplication of the normative value change by e^{-x}. The transformed change $e^{-x}dV_n/dx_n$ tends to zero at both extremes: on each tail, changing the cutoffs changes only the multiplier of the exponential,

whose exponent was identified above. The rogue value change also tends to zero at both extremes. Uniqueness for their common linear equation with these tail and matching conditions gives

$$\frac{dV_n(x;\alpha)}{dx_n} = e^{x-x_I}\frac{V_n'(x_I)}{V_r'(x_I)}\frac{dV_r(x;\alpha)}{dx_n}. \tag{B.8}$$

Movement of either stranger's own optimal quitting cutoff contributes no additional first-order boundary term, because her value there is zero.

Lastly, evaluate the normative value at the moving conjectured cutoff. The chain rule and equations (B.6) and (B.8) give

$$\begin{aligned}\frac{dV_n(x_n;\alpha)}{dx_n} &= V_n'(x_n) + \left.\frac{dV_n(x;\alpha)}{dx_n}\right|_{x=x_n} \\ &= e^{x_n-x_I}\frac{V_n'(x_I)}{V_r'(x_I)}\left[V_r'(x_n) + \left.\frac{dV_r(x;\alpha)}{dx_n}\right|_{x=x_n}\right] \\ &< 0.\end{aligned}$$

The multiplier is positive, and the bracket is negative by the bound just proved. This inequality holds at every zero of the normative consistency condition. Since every zero is crossed strictly downward, it cannot have two zeros. Hence there is exactly one consistent stranger pair for the fixed insider cutoff.

Before varying x_I, it remains to show that an insider facing any ordered stranger cutoffs $x_r < x_n$ optimally uses a cutoff. This is needed because indifference at a candidate x_I alone does not verify the insider's behavior at other reputations. Subtracting M_n from the insider's Bellman equation gives

$$\begin{aligned}0 = \frac{\psi^2}{2}(W-M_n)'' + \left(\Lambda + \frac{\psi^2}{2}\tanh(x/2)\right)(W-M_n)' - (\rho+\lambda H)(W-M_n) + \lambda(M_n-W)^+ \\ + \pi + \big(\rho + \lambda\mathbb{1}\{\alpha_n = q\}\big)F_n(1-p(x)) - \big(\rho+\lambda\mathbb{1}\{\alpha_r = q\}\big)G_n p(x).\end{aligned}$$

The value bounds make $W - M_n$ positive at sufficiently low reputations and negative at sufficiently high reputations. Suppose it becomes positive again after being negative. Consider the resulting bounded positive interval. At its left endpoint, $M_n = W \geq \pi/(\rho+\lambda)$, so the same lower bound on M_n holds throughout the interval. If its positive maximum occurred at or above x_r, the rogue would quit there, and the last line of the preceding equation would be at most

$$\pi - (\rho+\lambda)M_n \leq 0.$$

The Bellman equation would then make $(W - M_n)'' > 0$ at the maximum, a contradiction. At x_r, the same conclusion follows from the right-hand equation. Hence the positive maximum lies below x_r.

The preceding negative interval has an earlier negative minimum, also below x_r. Both strangers collaborate at these two reputations, so the last line reduces to $\pi - \rho M_n(x)$, a strictly decreasing function. The equation requires this function to be negative at the earlier negative minimum and positive at the later positive maximum, which is impossible. Thus the insider cannot return to collaboration after entering a whistleblowing region.

There can be no additional zero contacts. At a zero minimum followed by positive values, the later positive maximum must lie below x_r, as above. The zero minimum also lies below x_r and requires $\pi - \rho M_n \leq 0$, whereas the later positive maximum requires $\pi - \rho M_n > 0$, contradicting strict decrease.

A zero maximum preceded by negative values has an earlier negative minimum. If both reputations lie below x_r, the minimum requires $\pi - \rho M_n < 0$ and the later zero maximum requires $\pi - \rho M_n \geq 0$, again a contradiction. If the zero maximum is at or above x_r, the source is strictly negative there, also contradicting a maximum. This uses $W(x) > \pi/(\rho + \lambda)$ at every finite x: always collaborating delays termination beyond the stranger's first opportunity with positive probability, because both stranger types collaborate in a sufficiently low tail. At x_r, use the right-hand equation. Finally, an interval of zeros is impossible because the source is strictly decreasing between stranger cutoffs. Therefore $W - M_n$ has exactly one zero and the insider's best response is a cutoff.

I now let x_I vary. Shifting reputation and all three cutoffs by the same amount leaves the strangers' Bellman equations unchanged: the type-specific drift is constant, and the cutoff indicators are unchanged by this common shift. By uniqueness of the consistent stranger pair, $x_r - x_I$ and $x_n - x_I$ therefore remain constant as x_I varies,

$$x_\theta(x_I + \Delta) = x_\theta(x_I) + \Delta, \qquad \theta \in \{n, r\}. \tag{B.9}$$

Evaluate $W(x_I)$ against the consistent stranger pair associated with that same x_I. Here W is the insider's optimized value against that pair; the candidate cutoff x_I is not imposed on his own decision rule. Holding an insider strategy fixed in the translated coordinates gives conditional payoffs independent of x_I, so its expected payoff is affine in $p(x_I)$. Taking the supremum over insider strategies makes $W(x_I)$ convex in $p(x_I)$. Since the conditional payoffs are uniformly bounded, this supremum is also continuous. Since

$M_n(x_I)$ is affine in this probability, $W(x_I) - M_n(x_I)$ is also convex and continuous. The value bounds imply

$$\lim_{x_I \to -\infty} \big[W(x_I) - M_n(x_I)\big] \geq \frac{\pi}{\rho + \lambda} + F_n > 0,$$
$$\lim_{x_I \to +\infty} \big[W(x_I) - M_n(x_I)\big] \leq \frac{\pi - \rho G_n}{\rho + \lambda} < 0.$$

Consequently, this difference has exactly one zero: two zeros would, by convexity, force its right-endpoint value to be nonnegative. Because $p(x_I)$ is strictly increasing, the indifference condition $W(x_I) = M_n(x_I)$ uniquely determines x_I. Every cutoff equilibrium must satisfy this condition. Together with uniqueness of the consistent stranger pair for fixed x_I, this proves that there is at most one cutoff equilibrium. Existence also follows. At the unique zero, the associated consistent stranger pair is optimal against the candidate insider cutoff. The insider's optimized value is indifferent at that same reputation, and his best response to the ordered stranger pair has exactly one zero, as proved above. His optimal cutoff therefore equals the candidate x_I. All three decision rules are consequently optimal and consistent, so there is exactly one cutoff equilibrium.

With uniqueness established, what remains is to order the insider's cutoff relative to the strangers' cutoffs. These satisfy $x_r^* < x_n^*$. The position of x_I^*, however, depends not only on the payoffs, but also on the primitives of the information process. Define

$$\bar{L}(\mu) := \frac{\pi}{\lambda}\left[1 + \frac{\sqrt{\mu^2 + 2\psi^2(\rho + 2\lambda)} - \mu}{\sqrt{\mu^2 + 2\psi^2\rho} + \mu}\right],$$

The function $\bar{L}(\mu)$ is the critical exposure loss at which a stranger facing constant reputation drift μ is indifferent between quitting and collaborating at the insider's whistleblowing cutoff. It equals the baseline tolerable loss, π/λ, multiplied by a continuation factor that accounts for discounting and movements of reputation across the cutoff. This factor captures the possibility of remaining below the whistleblowing threshold or returning below it before termination. A higher drift makes the dangerous region easier to reach and harder to leave, reducing the loss the stranger is willing to tolerate. Consequently, $\bar{L}(\mu)$ is strictly decreasing in its argument. For a finite cutoff equilibrium, the following sufficient conditions determine the cutoff ordering:

(i) $x_I^* < x_r^* < x_n^*$ if $L_r < \bar{L}\,(\psi^2/2)$;

(ii) $x_r^* < x_I^* < x_n^*$ if $L_r > \bar{L}\,(\psi^2/2 - \lambda)$ and $L_n < \bar{L}\,(-\psi^2/2)$;

(iii) $x_r^* < x_n^* < x_I^*$ if $L_n > \bar{L}\,(-\psi^2/2 - \lambda)$.

To see this, fix an equilibrium (x_n^*, x_r^*, x_I^*) and translate the reputation so that the insider's cutoff is zero. First consider a stranger facing a constant reputation drift:

$$dX_t = \mu\, dt + \psi dB_t.$$

Let $U_\theta(x;\mu)$ denote the stranger's optimized value for this constant-drift process when the insider's cutoff is zero and the stranger's own opportunities still arrive at rate λ. Suppose the stranger's optimal cutoff also equals zero. This supposition identifies a critical loss, rather than imposing the stranger's cutoff for every loss. Below zero, both players collaborate. Above zero, each terminates at the next opportunity. The value function of the stranger must therefore take the form

$$U_\theta(x;\mu) = \begin{cases} \dfrac{\pi}{\rho}(1 - e^{\xi_1(\mu)x}) & \text{if } x \le 0, \\ \dfrac{\pi - \lambda L_\theta}{\rho + 2\lambda}(1 - e^{-\xi_2(\mu)x}) & \text{if } x > 0, \end{cases}$$

where

$$\xi_1(\mu) = \frac{\sqrt{\mu^2 + 2\psi^2\rho} - \mu}{\psi^2} > 0, \quad \xi_2(\mu) = \frac{\sqrt{\mu^2 + 2\psi^2(\rho + 2\lambda)} + \mu}{\psi^2} > 0.$$

Matching first derivatives at zero gives

$$\frac{\pi}{\rho}\xi_1(\mu) = \frac{\lambda L_\theta - \pi}{\rho + 2\lambda}\xi_2(\mu).$$

Solving for the loss yields,

$$L_\theta = \bar{L}(\mu).$$

The displayed piecewise formula applies at this critical loss; $U_\theta(x;\mu)$ continues to denote the optimized value at other losses. The value decreases strictly with the loss from the insider's whistleblow: at every finite reputation, there is positive probability of an insider accusation before the stranger's first opportunity, independently of her decision rule. Since the stranger's value decreases strictly with this loss, the critical loss determines which side of the insider's cutoff she quits on:

$$L_\theta < \bar{L}(\mu) \implies U_\theta(0;\mu) > 0 \implies x_\theta > x_I, \qquad L_\theta > \bar{L}(\mu) \implies U_\theta(0;\mu) < 0 \implies x_\theta < x_I,$$

Now return to the equilibrium. Because the rogue stranger quits weakly earlier than the normative stranger, the survival drift satisfies $-\lambda \le \Lambda(x) \le 0$. The two types' base

reputation drifts are $\mu_r = \psi^2/2$ and $\mu_n = -\psi^2/2$. Thus, coupling the processes with the same Brownian motion gives

$$x + (\mu_\theta - \lambda)t + \psi B_t \leq X_t \leq x + \mu_\theta t + \psi B_t.$$

A lower reputation path delays the insider's whistleblowing and raises the stranger's continuation value. Comparing these optimal stopping problems yields

$$U_\theta(x; \mu_\theta) \leq V_\theta(x) \leq U_\theta(x; \mu_\theta - \lambda).$$

Thus,

$$L_\theta < \bar{L}(\mu_\theta) \implies x^*_\theta > x^*_I$$

and

$$L_\theta > \bar{L}(\mu_\theta - \lambda) \implies x^*_\theta < x^*_I$$

Applying these implications to each type establishes the three stated sufficient conditions for the cutoff orderings.

B.2. Partnership Dissolution

This subsection establishes each of the four claims of Proposition 4 in turn. Fix an equilibrium with finite cutoffs. By Proposition 2, $x^*_r < x^*_n$. Conditional on the stranger's type

$$dX_t = \mu_\theta(X_t)dt + \psi dB_t, \tag{B.10}$$

where

$$\mu_\theta(x) = \frac{\psi^2}{2}(\mathbb{1}\{\theta = r\} - \mathbb{1}\{\theta = n\}) - \lambda(\mathbb{1}\{x^*_r < x < x^*_n\}).$$

Throughout the proof, X_t denotes the solution of (B.10) for all $t \geq 0$, driven by the Brownian motion independently of the opportunity process; it agrees with equilibrium reputation until the terminal time τ,

$$\tau_I = \inf\{T^I_k : X_{T^I_k} > x^*_I\}, \quad \tau_\theta = \inf\{T^S_k : X_{T^S_k} > x^*_\theta\}, \quad \tau = \min\{\tau_I, \tau_\theta\}.$$

Claim (i): Only partnerships with normative strangers can last forever. Once the entire reputation path is fixed, a partnership last forever if

$$X_{T^I_k} \leq x^*_I \quad \text{for every } k, \quad X_{T^S_k} \leq x^*_\theta \quad \text{for every } k.$$

Thus, the insider must have no opportunity during the times when $X_t > x_I^*$, and the stranger must have no opportunity during the times when $X_t > x_\theta^*$. Independence of the two opportunity process gives

$$\mathbb{P}_\theta^x(\tau = \infty) = \mathbb{E}_\theta^x \left[\exp\left(-\lambda \int_0^\infty \left(\mathbb{1}\{X_t > x_I^*\} + \mathbb{1}\{X_t > x_\theta^*\} \right) dt \right) \right]. \tag{B.11}$$

For a normative stranger, (B.10) implies

$$X_t \leq x - \frac{\psi^2}{2} t + \psi B_t \to -\infty \quad \textit{almots surely.}$$

Since both stopping cutoffs are finite, the process eventually remains below both. The integral in (B.11) is therefore finite almost surely. Its exponential is strictly positive, so

$$\mathbb{P}_\theta^x(\tau = \infty) > 0, \quad \omega_n(x) + q_n(x) < 1.$$

For a rogue stranger, I show that $X_t \to \infty$. Take two levels $x_1 < x < x_2$. Before either boundary is reached, the probability of eventually reaching x_2 first can be reassessed using the current reputation. The average of these future reassessed probabilities equals the current probability, because they point the same event. If a boundary has already been reached, the reassessed probability is zero at x_1 and one at x_2.

Since this conditional probability is a martingale, applying Itô's formula and taking conditional expectations gives

$$\frac{\psi^2}{2} \frac{d^2}{dx^2} \mathbb{P}_r^x(\text{reach } x_2 \text{ before } x_1) + \mu_r(x) \frac{d}{dx} \mathbb{P}_r^x(\text{reach } x_2 \text{ before } x_1) = 0.$$

Solving with boundary values zero at x_1 and one at x_2 gives

$$\mathbb{P}_r^x(\text{reach } x_2 \text{ before } x_1) = \frac{\displaystyle\int_{x_1}^{x} \exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2} \, dz \right] dy}{\displaystyle\int_{x_1}^{x_2} \exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2} \, dz \right] dy}. \tag{B.12}$$

Bounded drift and $\psi > 0$ ensure that the diffusion exist every finite interval almost

surely. For the drift in (B.10),

$$\exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2}\,dz\right] = \exp\left[-y + \frac{2\lambda}{\psi^2}\int_0^y \mathbb{1}\{x_r^* < z < x_n^*\}\,dz\right].$$

Since

$$\left|\int_0^y \mathbf{1}_{\{x_r^*<z<x_n^*\}}\,dz\right| \le x_n^* - x_r^*,$$

this integrand is bounded above and below by positive multiples of e^{-z}.

Sending $x_1 \to -\infty$ in (B.12), the numerator and denominator diverge while their difference remains finite. Their ratio tends to one. Hence every upper level is reached almost surely. Similarly starting at $x_2 > x_1$, sending the upper boundary to infinity gives

$$\mathbb{P}_r^{x_2}(\text{ever reach } x_1) = \frac{\displaystyle\int_{x_2}^{\infty} \exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2}\,dz\right] dy}{\displaystyle\int_{x_1}^{\infty} \exp\left[-\int_0^y \frac{2\mu_r(z)}{\psi^2}\,dz\right] dy} \to 0 \qquad (x_2 \to \infty). \tag{B.13}$$

The denominator is positive and finite, while the numerator tends to zero.

Fix a lower level x_1. By the strong Markov property, after the reputation first reaches a higher level x_2, the probability of subsequently returning to x_1 is give by (B.13). First visits to successively higher integer levels occur almost surely and in increasing order. The events of returning to x_1 after these visits form a decreasing sequence whose probabilities tend to zero. Almost surely, one such visit is therefore followed by no return to x_1. Continuity then keeps the process above x_1 forever. Applying this argument to every integer lower level proves

$$X_t \to \infty \quad \text{as } t \to \infty, \qquad \mathbb{P}_r^x\text{-almost surely.}$$

The process eventually remains above both stopping cutoffs, making the integral inside the exponential in (B.11) infinite. Consequently,

$$\omega_r(x) + q_r(x) = 1.$$

Claim (ii): The partner with the lower cutoff is more likely to end the partnership. While the partnership last, an insider opportunity arrive at rate λ and causes a whistleblow exactly when $X_t > x_I^*$. Adding these arrival probabilities over time gives,

$$\begin{aligned}\omega_\theta(x) &= \lambda \int_0^\infty \mathbb{P}_\theta^x\big(t < \tau,\ X_t > x_I^*\big)\, dt \\ &= \lambda\, \mathbb{E}_\theta^x \left[\int_0^\tau \mathbb{1}\{X_t > x_I^*\}\, dt \right]. \end{aligned} \tag{B.14}$$

The same argument for stranger opportunities gives

$$q_\theta(x) = \lambda\, \mathbb{E}_\theta^x \left[\int_0^\tau \mathbb{1}\{X_t > x_\theta^*\}\, dt \right]. \tag{B.15}$$

These equations hold first on finite horizons and extend to the infinite horizon by monotone convergence. To compare the dissolution probabilities, first consider the case $x_\theta^* < x_I^*$. Subtracting (B.14) from (B.15) yields

$$q_\theta(x) - \omega_\theta(x) = \lambda\, \mathbb{E}_\theta^x \left[\int_0^\tau \mathbb{1}\{x_\theta^* < X_t \le x_I^*\}\, dt \right] > 0. \tag{B.16}$$

The inequality is strict. Fix a finite horizon T. Since the drift is bounded, Girsanov's theorem implies that, on $[0, T]$, the law of X is equivalent to that of $(x + \psi B_t)_{t\in[0,T]}$. Hence, from any initial reputation x, with positive probability the process enters $(x_\theta^*, x_I^*]$ before time $T/2$ and remains there for a positive length of time before T. Independently of the Brownian motion, with positive probability neither opportunity process arrives before T, so $\tau > T$. On the intersection of these events, the occupation time in (B.16) is strictly positive, and therefore so is its expectation.

If instead $x_I^* < x_\theta^*$, the same argument, with the roles of the cutoffs reversed, gives

$$\omega_\theta(x) - q_\theta(x) = \lambda\, \mathbb{E}_\theta^x \left[\int_0^\tau \mathbb{1}\{x_I^* < X_t \le x_\theta^*\}\, dt \right] > 0.$$

Finally, if $x_\theta^* = x_I^*$, the indicators in (B.15) and (B.14) coincide, so $\omega_\theta(x) = q_\theta(x)$. This condition cannot hold for both types simultaneously because $x_r^* < x_n^*$.

Claim (iii): However low the initial suspicion, a normative stranger is falsely accused with positive probability. By (B.14), the probability that the insider dissolves the relationship when the agent is normal is

$$\omega_n(x) = \lambda\, \mathbb{E}_n^x \left[\int_0^\tau \mathbb{1}\{X_t > x_I^*\}\, dt \right].$$

The positive-occupation argument from Claim (ii), applied to the region (x_I^*, ∞) instead of the interval between the cutoffs, shows that the expected occupation time is strictly positive. Hence $\omega_n(x) > 0$ for every finite initial reputation x.

Claim (iv): Partnership value strictly decreases with suspicion. Fix the stranger's type θ, and let X^1 and X^2 denote reputation processes started at $x_1 < x_2$, driven by the same Brownian motion and the same opportunity processes. Their paths are continuous and cannot cross: if they ever meet, pathwise uniqueness for the reputation equation forces them to coincide from then on. Hence $X_t^1 \le X_t^2$ for all t.

Termination occurs when an opportunity arrives while reputation lies above the relevant cutoff. Since X^2 is above any given cutoff whenever X^1 is, every opportunity that ends the partnership started at x_1 also ends the partnership started at x_2. Thus $\tau^1 \ge \tau^2$ almost surely: the partnership started at x_2 terminates weakly earlier. Because the flow value is nonnegative (see below), a shorter partnership is worth weakly less, so v_θ is nonincreasing.

The value v_θ satisfies

$$\frac{\psi^2}{2} v_\theta''(x) + \mu_\theta(x)\, v_\theta'(x) - \Big(\rho + \lambda \mathbb{1}\{x > x_I^*\} + \lambda \mathbb{1}\{x > x_\theta^*\}\Big) v_\theta(x) + \pi = 0. \tag{B.17}$$

Besides discounting at rate ρ, the value is lost at rate λ whenever reputation lies above a player's cutoff, since an arriving opportunity then ends the partnership. The function v_θ is smooth between cutoffs. Integrating (B.17) over a shrinking interval around a cutoff shows that v_θ' is continuous there, so v_θ is continuously differentiable, while v_θ'' may jump. Since the total termination rate lies between 0 and 2λ,

$$\frac{\pi}{\rho + 2\lambda} \le v_\theta(x) \le \frac{\pi}{\rho}. \tag{B.18}$$

It remains to show that v_θ is strictly decreasing. By the coupling argument, $v_\theta' \le 0$, so any zero of v_θ' is a maximum of v_θ'.

Zeros between cutoffs. Differentiating (B.17) between cutoffs gives

$$\frac{\psi^2}{2} v_\theta''' + \mu_\theta \, v_\theta'' - \Big(\rho + \lambda \mathbb{1}\{x > x_I^*\} + \lambda \mathbb{1}\{x > x_\theta^*\}\Big) v_\theta' = 0, \tag{B.19}$$

a linear second-order equation in v_θ'. If v_θ' vanishes at an interior point, that point is a maximum, so v_θ'' vanishes there as well. With both initial values equal to zero, uniqueness for (B.19) forces $v_\theta' \equiv 0$ on the whole interval, and by continuity at its endpoints.

Zeros at a cutoff. Let z be a cutoff with $v_\theta'(z) = 0$. Subtracting the left-hand limit of (B.17) at z from the right-hand limit, the drift term drops out because $v_\theta'(z) = 0$, and

$$\frac{\psi^2}{2}\Big(v_\theta''(z+) - v_\theta''(z-)\Big) = \lambda \, v_\theta(z)\Big(\mathbb{1}\{z = x_I^*\} + \mathbb{1}\{z = x_\theta^*\}\Big). \tag{B.20}$$

Since v_θ' is nonpositive and attains 0 at z, we have $v_\theta''(z-) \geq 0 \geq v_\theta''(z+)$, so the left-hand side is nonpositive. If z is a stopping cutoff ($z \in \{x_I^*, x_\theta^*\}$), the right-hand side is strictly positive by (B.18), a contradiction. If only the drift changes at z, (B.20) makes v_θ'' continuous at z, and the one-sided inequalities force $v_\theta''(z) = 0$. Applying the previous step on the adjacent interval, the zero then propagates across it.

Combining the two steps, any zero of v_θ' propagates from interval to interval until it reaches a stopping cutoff, where it yields a contradiction. Hence

$$v_\theta'(x) < 0 \qquad \text{for every finite } x.$$

Finally, compare v_n and v_r. Let τ be the termination time of the partnership with the normative type. Applying Dynkin's formula to $e^{-\rho t} v_r(X_t)$ along the normative reputation process, whose drift is μ_n, gives

$$\mathbb{E}_n^x\big[e^{-\rho\tau} v_r(X_\tau)\big] - v_r(x) = \mathbb{E}_n^x\left[\int_0^\tau e^{-\rho t}\Big(\frac{\psi^2}{2} v_r''(X_t) + \mu_n(X_t)\, v_r'(X_t) - \rho \, v_r(X_t)\Big) dt\right]. \tag{B.21}$$

Dynkin's formula applies because v_r is C^1 with a piecewise continuous second derivative. It is applied first at $\tau \wedge T$, and then $T \to \infty$; the terminal term vanishes on $\{\tau = \infty\}$ because v_r is bounded and $\rho > 0$. By the arrival argument behind (B.15)–(B.14), with the terminal amount $v_r(X_t)$ in place of one, the expectation on the left-hand side of

(B.21) also equals

$$\mathbb{E}_n^x\big[e^{-\rho\tau}v_r(X_\tau)\,\mathbb{1}\{\tau<\infty\}\big]=\lambda\,\mathbb{E}_n^x\left[\int_0^\tau e^{-\rho t}\Big(\mathbb{1}\{X_t>x_I^*\}+\mathbb{1}\{X_t>x_n^*\}\Big)v_r(X_t)\,dt\right]. \tag{B.22}$$

Substitute (B.22) into (B.21) and eliminate $\frac{\psi^2}{2}v_r''$ via the rogue version of (B.17). Since $\mu_r-\mu_n=\psi^2$, $v_n(x)=\mathbb{E}_n^x\big[\int_0^\tau e^{-\rho t}\pi\,dt\big]$, and $x_r^*<x_n^*$, this yields

$$v_n(x)-v_r(x)=\mathbb{E}_n^x\left[\int_0^\tau e^{-\rho t}\Big(\lambda\,\mathbb{1}\{x_r^*<X_t\le x_n^*\}\,v_r(X_t)-\psi^2v_r'(X_t)\Big)dt\right].$$

The integrand is strictly positive, because $v_r>0$ by (B.18) and $v_r'<0$. Since $\tau>0$ almost surely, $v_n(x)>v_r(x)$ for every finite x.

B.3. Comparative Statics

The invariance property in (B.9) provides a convenient basis for the comparative statics. For fixed parameters, changing the candidate insider cutoff shifts both consistent stranger cutoffs by the same amount, leaving the distances between cutoffs unchanged.

Consider first the insider's whistleblowing gain, G_n. Holding all other parameters fixed, increasing this gain lowers all three equilibrium cutoffs by the same amount:

$$\frac{\partial x_I^*}{\partial G_n}=\frac{\partial x_n^*}{\partial G_n}=\frac{\partial x_r^*}{\partial G_n}<0. \tag{B.23}$$

To see this, fix a x_I. The strangers' Bellman equations do not involve the insider's gain. The previous translation argument therefore implies that the distance between the stranger cutoffs and the insider cutoff are independent of that gain:

$$\frac{\partial}{\partial G_n}(x_n-x_I)=\frac{\partial}{\partial G_n}(x_r-x_I)=0.$$

Next, consider insider indifference:

$$W(x_I)=M_n(x_I)=p(x_I)G_n-[1-p(x_I)]F_n.$$

Holding the candidate cutoff and its consistent stranger pair fixed, the immediate whistleblowing payoff increase at a rate $p(x_I)$. The continuation value, however, increases more slowly. Whistleblowing cannot occur before the insider's first decision opportu-

nity, whose discount factor is $\lambda/(\rho+\lambda)$. Therefore, wherever the derivative exists,

$$0 \leq \frac{\partial W(x_I)}{\partial G_n} \leq \frac{\lambda}{\rho+\lambda} p(x_I).$$

This bounds holds for every insider strategy and hence also for the optimal cutoff. Thus, increasing G_n strictly reduces the difference between continuation and immediate whistleblowing:

$$\frac{\partial}{\partial G_n}[W(x_I) - M_n(x_I)] < 0.$$

By the previous uniqueness argument, this difference cross zero strictly downward as the candidate insider cutoff increases. Differentiating the equilibrium indifference condition therefore gives

$$\frac{\partial x_I^*}{\partial G_n} = \frac{p(x_I^*) - \left.\dfrac{\partial W(x_I)}{\partial G_n}\right|_{x_I = x_I^*}}{\left.\dfrac{d}{dx_I}[W(x_I) - M_n(x_I)]\right|_{x_I = x_I^*}} < 0.$$

The numerator is strictly positive, and the denominator strictly negative. Lastly, since the distance between cutoffs are independent of the gain yields (B.23).

C. Omitted Proofs

Proof of Proposition 1. The law of motion is derived for an arbitrary player k, then specialized to player i and her partner j separately.

By Bayes' rule, the posterior odds ratio for player k given the event $\{\tau > t\}$ decomposes as

$$\frac{\mathbb{P}(\theta_k = r \mid \mathcal{Z}_t, \{\tau > t\})}{\mathbb{P}(\theta_k = n \mid \mathcal{Z}_t, \{\tau > t\})} = \frac{\mathbb{P}(\theta_k = r \mid \mathcal{Z}_t)}{\mathbb{P}(\theta_k = n \mid \mathcal{Z}_t)} \cdot \frac{\mathbb{P}(\{\tau > t\} \mid \mathcal{Z}_t, \theta_k = r)}{\mathbb{P}(\{\tau > t\} \mid \mathcal{Z}_t, \theta_k = n)},$$

where the first term on the right-hand side is the signal contribution and the second term is the survival contribution. To compute the signal contribution, recall that Z^k satisfies $dZ_t^k = \mu(\theta_k)\,dt + \sigma\,dB_t^k$. Let $\phi_r(\cdot)$ and $\phi_n(\cdot)$ denote the corresponding Gaussian densities of Z_t^k under types r and n, respectively. Since the drift is constant, Z_t^k is a sufficient statistic for θ_k: by Girsanov's theorem, the likelihood ratio of the path $(Z_s^k)_{s\leq t}$ depends on it only through Z_t^k and equals the ratio of these densities,

$$\frac{\phi_r(Z_t^k)}{\phi_n(Z_t^k)} = \exp\left(\frac{\mu(r)-\mu(n)}{\sigma^2} Z_t^k - \frac{(\mu(r)+\mu(n))(\mu(r)-\mu(n))}{2\sigma^2} t\right).$$

Thus, given the prior p_0

$$\frac{\mathbb{P}(\theta_k = r \mid \mathcal{Z}_t)}{\mathbb{P}(\theta_k = n \mid \mathcal{Z}_t)} = \frac{p_0}{1-p_0}\frac{\phi_r(Z_t^k)}{\phi_n(Z_t^k)}.$$

Taking logs and differentiating yields the signal contribution to dX_t^k:

$$\frac{\mu(r)-\mu(n)}{\sigma^2}dZ_t^k - \frac{(\mu(r)+\mu(n))(\mu(r)-\mu(n))}{2\sigma^2}dt. \tag{C.24}$$

Player i knows her type drift $\mu(\theta_i)$, and substituting it into dZ_t^i in (C.24) yields

$$\frac{\mu(r)-\mu(n)}{\sigma^2}\mu(\theta_i)\,dt - \frac{(\mu(r)+\mu(n))(\mu(r)-\mu(n))}{2\sigma^2}\,dt + \frac{\mu(r)-\mu(n)}{\sigma}dB_t^i.$$

Using $\psi = (\mu(r)-\mu(n))/\sigma$, this simplifies to

$$\frac{\psi^2}{2}\big(\mathbb{1}\{\theta_i = r\} - \mathbb{1}\{\theta_i = n\}\big)\,dt + \psi\,dB_t^i.$$

Player j's type is unknown to player i, so the drift of Z^j is replaced by its posterior mean given the public information, $\mathbb{E}[\mu(\theta_j) \mid \mathcal{Z}_t, \{\tau > t\}] = \mu(r)P_t^j + \mu(n)(1-P_t^j)$. The *innovation* process associated with Z^j is obtained by subtracting this conditional drift from the observed signal,

$$d\widetilde{B}_t^j := \frac{1}{\sigma}\left(dZ_t^j - \mathbb{E}[\mu(\theta_j) \mid \mathcal{Z}_t, \{\tau > t\}]\,dt\right).$$

By standard filtering arguments, $\widetilde{B}^j$ is a Brownian motion with respect to the public information. Player i's private information, her type and her opportunities, is independent of θ_j given the public information, so conditioning on it leaves the posterior mean unchanged, and $\widetilde{B}^j$ is a Brownian motion from her perspective as well. Substituting the resulting innovation representation into (C.24) yields

$$\psi^2\left(P_t^j - \frac{1}{2}\right)dt + \psi\,d\widetilde{B}_t^j.$$

Equivalently, the drift can be written in log-odds form as

$$\frac{\psi^2}{2}\tanh\left(\frac{X_t^j}{2}\right)\,dt + \psi\,d\widetilde{B}_t^j.$$

Next, consider the information conveyed by the fact that the game has survived up to

time t. Unlike the signal process, survival is endogenous to players' stopping behavior and therefore strategy-dependent. Let α and β denote the strategies of players i and j, respectively, and let τ_i and τ_j denote their respective stopping times. Since the game ends when either player stops, $\tau = \tau_i \wedge \tau_j$, and hence

$$\{\tau > t\} = \{\tau_i > t\} \cap \{\tau_j > t\}.$$

Consider first the information conveyed about θ_i. Conditional on the realized public state path, player j's stopping behavior depends on her own type, but not directly on θ_i. Player j's survival, $\{\tau_j > t\}$, enters with the same likelihood under $\theta_i = r$ and $\theta_i = n$ and cancels from the likelihood ratio between the two types. Thus, the survival contribution to beliefs about θ_i is determined entirely by player i's own survival.

Conditional on θ_i, player i stops at the first opportunity at which $(X_t^i, X_t^j) \notin \mathcal{C}_{\theta_i}(\alpha)$. Thus, conditional on the public state at time t, the probability that player i has not stopped by time t is

$$\mathbb{P}(\{\tau_i > t\} \mid \mathcal{Z}_t, \theta_i) = \exp\left(-\lambda \int_0^t \mathbb{1}\{(X_s^i, X_s^j) \notin \mathcal{C}_{\theta_i}(\alpha)\} ds\right).$$

It follows that the likelihood ratio associated with player i's survival is

$$\frac{\mathbb{P}(\{\tau_i > t\} \mid \mathcal{Z}_t, \theta_i = r)}{\mathbb{P}(\{\tau_i > t\} \mid \mathcal{Z}_t, \theta_i = n)} = \exp\left(\lambda \int_0^t \left[\mathbb{1}\{(X_s^i, X_s^j) \notin \mathcal{C}_n(\alpha)\} - \mathbb{1}\{(X_s^i, X_s^j) \notin \mathcal{C}_r(\alpha)\}\right] ds\right).$$

Taking logs, survival contributes $+\lambda$ to the drift of the log-odds at states in which only rogue types collaborate, i.e., $\mathcal{I}_r(\alpha)$. Conversely, it contributes $-\lambda$ at states in which only normative types collaborate, i.e., $\mathcal{I}_n(\alpha)$. Hence, survival contributes the drift $\Lambda(x, y; \alpha)$ to dX_t^i. An analogous argument for player j yields the drift $\Lambda(y, x; \beta)$ in dX_t^j. Finally, because the likelihood contributions of the signal and survival multiply under Bayes' rule, their log-likelihood contributions are additive. Combining the two therefore yields the stated log-odds dynamics. □

Proof of Lemma A.1. The process (X_t^i, X_t^j) characterized in Proposition 1 has infinitesimal generator

$$\mathcal{L}_\theta = \frac{\psi^2}{2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + \mu_i \frac{\partial}{\partial x} + \mu_j \frac{\partial}{\partial y},$$

where

$$\mu_i = \Lambda(x, y; \alpha) + \tfrac{\psi^2}{2}\big(\mathbb{1}\{\theta_i = r\} - \mathbb{1}\{\theta_i = n\}\big), \tag{C.25}$$

$$\mu_j = \Lambda(y, x; \alpha) + \tfrac{\psi^2}{2}\tanh(y/2). \tag{C.26}$$

The drifts μ_i and μ_j are bounded and Borel measurable : $\Lambda(\cdot)$ is a linear combination of indicator functions, while the remaining drift terms are bounded and continuous. Since the diffusion matrix is constant and $\psi > 0$, $\mathcal{L}_\theta$ is uniformly elliptic.

Existence and uniqueness of a strong solution follow from Theorem 1 of Veretennikov (1981). Moreover, by Stroock and Varadhan (2007), the martingale problem associated with $\mathcal{L}_\theta$ is well-posed (Theorem 7.2.1), the resulting process is strong Markov (Theorem 6.2.2), and its transition semigroup is strong Feller (Lemma 9.2.2). □

Proof of Proposition 5. The proof first reformulates the problem in opportunity time and shows that the corresponding Bellman operator is a contraction. It then connects this discrete-time recursion to the original continuous-time problem by iterating over successive opportunities. This identifies the unique fixed point of the Bellman operator with the value function and yields the myopic characterization of optimal behavior.

Fix a strategy α and a type θ. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a bounded Borel-measurable function representing a candidate continuation value. At a stopping opportunity, the player chooses between stopping for $M_\theta(y)$ and continuing with value $f(x, y)$. Accordingly, define

$$\Upsilon_\theta(x, y; f) := \max\{M_\theta(y),\ f(x, y)\}.$$

Similarly, when the partner follows strategy α, let $\Gamma_\theta(x, y; f)$ denote the player's value at a partner's opportunity, as defined in Appendix A.1.

Let $T := \min\{T_1^i, T_1^j\} \sim \text{Exp}(2\lambda)$ denote the first opportunity time of either player. Conditional on $T = t$, the opportunity belongs to each player with probability $1/2$. The value admits a decomposition into the discounted flow payoff accrued up to T and the discounted value at T, which depends on which player receives the opportunity. Integrating this decomposition against the density of T yields the Bellman operator

$$\Phi_\theta^\alpha[f](x, y) := \int_0^\infty 2\lambda e^{-2\lambda t} \left(\int_0^t e^{-\rho s}\pi\, ds + \frac{e^{-\rho t}}{2}\, \mathbb{E}_{x,y}\big[\Upsilon_\theta(X_t^i, X_t^j; f) + \Gamma_\theta(X_t^i, X_t^j; f)\big] \right) dt.$$

The operator Φ_θ^α maps $C_b^0(\mathbb{R}^2)$ into itself. Boundedness follows directly from bounded payoffs and continuation values. For continuity, the flow-payoff term $\int_0^\infty 2\lambda e^{-2\lambda t} \int_0^t e^{-\rho s}\pi\, ds\, dt$

is constant in (x, y). The remaining term has the form

$$\int_0^\infty \lambda e^{-(2\lambda+\rho)t}\, \mathbb{E}_{x,y}\big[\Upsilon_\theta(X_t^i, X_t^j; f) + \Gamma_\theta(X_t^i, X_t^j; f)\big]\, dt. \tag{C.27}$$

The function $\Upsilon_\theta(\cdot; f) + \Gamma_\theta(\cdot; f)$ is bounded and Borel measurable. By the strong Feller property in Lemma A.1, the map $(x, y) \mapsto \mathbb{E}_{x,y}[\Upsilon_\theta(X_t^i, X_t^j; f)+\Gamma_\theta(X_t^i, X_t^j; f)]$ is continuous for each $t > 0$. Since the expectation is uniformly bounded, dominated convergence applies using a constant multiple of $\lambda e^{-(2\lambda+\rho)t}$ as an integrable bound, and therefore implies continuity in (x, y) after integration over t. Hence $\Phi_\theta^\alpha[f] \in C_b^0(\mathbb{R}^2)$.

To show Φ_θ^α is a contraction, take $f, g \in C_b^0(\mathbb{R}^2)$ then,

$$|\Upsilon_\theta(x, y; f) - \Upsilon_\theta(x, y; g)| \le |f(x, y) - g(x, y)|.$$

This follows from $|\max\{a, b\} - \max\{a, c\}| \le |b - c|$. For Γ_θ, the terms involving $u(\cdot)$ do not depend on f or g and cancel. The remaining terms, weighted by $p(y)\mathbb{1}\{(y, x) \in \mathcal{C}_r(\alpha)\}$ and $(1 - p(y))\mathbb{1}\{(y, x) \in \mathcal{C}_n(\alpha)\}$, have nonnegative coefficients that sum to at most one. Therefore,

$$|\Gamma_\theta(x, y; f) - \Gamma_\theta(x, y; g)| \le |f(x, y) - g(x, y)|.$$

Using these two bounds in (C.27) yields

$$\begin{aligned}
|\Phi_\theta^\alpha[f](x, y) - \Phi_\theta^\alpha[g](x, y)| &\le \int_0^\infty \lambda e^{-(2\lambda+\rho)t} \cdot 2\mathbb{E}_{x,y}\big[|f(X_t^i, X_t^j) - g(X_t^i, X_t^j)|\big]\, dt \\
&\le \|f - g\|_\infty \int_0^\infty 2\lambda e^{-(2\lambda+\rho)t}\, dt \\
&= \frac{2\lambda}{2\lambda + \rho}\, \|f - g\|_\infty.
\end{aligned}$$

Since $\rho > 0$, Φ_θ^α is a contraction on $C_b^0(\mathbb{R}^2)$ with modulus $\frac{2\lambda}{2\lambda+\rho} < 1$. By the Banach fixed-point theorem, Φ_θ^α has a unique fixed point $V_\theta(\cdot; \alpha) \in C_b^0(\mathbb{R}^2)$, establishing boundedness and continuity of the value function.

It remains to identify the fixed point with the value function defined in (3) and to derive the myopic characterization. To this end, construct a candidate optimizer φ_θ^*, show that it attains V_θ, and verify that no alternative decision rule yields a higher payoff.

Define the decision rule

$$\varphi^*_\theta(x,y) := \begin{cases} c & \text{if } V_\theta(x,y;\alpha) > M_\theta(y), \\ m_\theta(y) & \text{otherwise.} \end{cases}$$

For any measurable decision rule φ and any $f \in C_b^0(\mathbb{R}^2)$, define

$$\Upsilon^\varphi_\theta(x,y;f) := \mathbb{1}\{\varphi(x,y) = c\}\, f(x,y) + \mathbb{1}\{\varphi(x,y) \neq c\}\, \overline{u}(\varphi(x,y);\theta,y),$$

and let Φ^φ_θ denote the operator obtained from Φ^α_θ by replacing Υ_θ with Υ^φ_θ. Note that $\Gamma_\theta(x,y;f)$ depends only on α, not on φ, so Φ^α_θ and Φ^φ_θ differ only through their Υ-terms. Since $\Upsilon_\theta(x,y;f) = \max\{M_\theta(y), f(x,y)\}$ dominates any single choice among stopping or continuing, while Υ^φ_θ commits to the choice prescribed by φ,

$$\Upsilon^\varphi_\theta(x,y;f) \le \Upsilon_\theta(x,y;f) \qquad \text{for every } \varphi, \tag{C.28}$$

with equality when $\varphi = \varphi^*$. Consequently $\Phi^{\varphi^*}_\theta[V_\theta] = \Phi^\alpha_\theta[V_\theta] = V_\theta$. The same contraction argument used for Φ^α_θ applies verbatim to Φ^φ_θ, hence V_θ is the unique fixed point of $\Phi^{\varphi^*}_\theta$ in $C_b^0(\mathbb{R}^2)$.

The operator Φ^φ_θ integrates against the first opportunity time, whereas J_θ runs until termination. Bridging the two requires re-expressing Φ^φ_θ along the sequence of opportunity times. Let $T_1 < T_2 < \cdots$ denote the ordered opportunity times of either player, so that T_k is the k-th arrival of a Poisson process of rate 2λ. Given φ and α, let τ be the termination time of the game: at each T_k, whoever has the opportunity consults their decision rule: φ for player i; α_{θ_j} for j. Let Ψ^φ_θ denote the expected terminal payoff accrued at τ. Unpacking the definition of Φ^φ_θ yields the one-step representation:

$$\Phi^\varphi_\theta[f](x,y) = \mathbb{E}^{\alpha,\varphi}_{x,y}\left[\int_0^{\tau\wedge T_1} e^{-\rho s}\pi\, ds + e^{-\rho\tau}\Psi^\varphi_\theta \mathbb{1}\{\tau \le T_1\} + e^{-\rho T_1} f(X^i_{T_1}, X^j_{T_1})\mathbb{1}\{\tau > T_1\}\right].$$

The claim is that iterating k times gives

$$\begin{aligned}(\Phi^\varphi_\theta)^k[f](x,y) = \mathbb{E}^{\alpha,\varphi}_{x,y}\Bigg[\int_0^{\tau\wedge T_k} e^{-\rho s}\pi\, ds + e^{-\rho\tau}\Psi^\varphi_\theta \mathbb{1}\{\tau \le T_k\} \\ + e^{-\rho T_k} f(X^i_{T_k}, X^j_{T_k})\mathbb{1}\{\tau > T_k\}\Bigg].\end{aligned} \tag{C.29}$$

Assume (C.29) holds for k. Applying Φ_θ^φ once more replaces $f(X^i_{T_k}, X^j_{T_k})$ on the event $\{\tau > T_k\}$ by $\Phi_\theta^\varphi[f](X^i_{T_k}, X^j_{T_k})$. By the strong Markov property at T_k, this equals

$$\mathbb{E}^{\alpha,\varphi}\left[\int_0^{\tau'\wedge T_1'} e^{-\rho s}\pi\,ds + e^{-\rho\tau'}\Psi_\theta^\varphi \mathbb{1}\{\tau' \le T_1'\} + e^{-\rho T_1'} f(X^i_{T_1'}, X^j_{T_1'})\,\mathbb{1}\{\tau' > T_1'\}\right],$$

where τ' and T_1' are the termination time and first opportunity time of the game restarted at T_k from state $(X^i_{T_k}, X^j_{T_k})$. Re-aligning to the original time, these correspond to τ and T_{k+1}. Combining with the part of the k-step formula already accrued on $\{\tau \le T_k\}$ yields the $(k+1)$-step formula, completing the induction.

Attainment. Apply (C.29) with $f = V_\theta$ and $\varphi = \varphi^*$. The left-hand side of (C.29) equals V_θ for every k, since V_θ is the fixed point of $\Phi_\theta^{\varphi^*}$. For the right-hand side, as $k \to \infty$, $T_k \to \infty$ a.s., so $\tau \wedge T_k \uparrow \tau$ and $\mathbb{1}\{\tau \le T_k\} \uparrow \mathbb{1}\{\tau < \infty\}$ a.s. Moreover,

$$|e^{-\rho T_k} V_\theta(X^i_{T_k}, X^j_{T_k})| \le \|V_\theta\|_\infty e^{-\rho T_k} \to 0 \quad \text{a.s.},$$

and the flow term is uniformly bounded by π/ρ. Dominated convergence then yields

$$V_\theta(x,y;\alpha) = \mathbb{E}_{x,y}^{\alpha,\varphi^*}\left[\int_0^{\tau} e^{-\rho s}\pi\,ds + e^{-\rho\tau}\Psi_\theta^{\varphi^*}\mathbb{1}\{\tau < \infty\}\right] = J_\theta(x,y;\varphi_\theta^*,\alpha).$$

Optimality. For any measurable φ, (C.28) gives $\Phi_\theta^\varphi[V_\theta] \le \Phi_\theta^\alpha[V_\theta] = V_\theta$. Since both Υ_θ^φ and Γ_θ are monotone in f, the operator Φ_θ^φ is monotone, so iterating yields $(\Phi_\theta^\varphi)^k[V_\theta] \le V_\theta$ for every k. Applying (C.29) to the left-hand side and passing to the limit $k \to \infty$ by dominated convergence as above,

$$J_\theta(x,y;\varphi,\alpha) \le V_\theta(x,y;\alpha).$$

The attainment and optimality results together give $V_\theta(x,y;\alpha) = \sup_\varphi J_\theta(x,y;\varphi,\alpha)$ with the supremum attained by φ_θ^*. By construction, φ_θ^* is myopically optimal and has collaboration region

$$\mathcal{C}_\theta = \{(x,y) \in \mathbb{R}^2 : V_\theta(x,y;\alpha) > M_\theta(y)\}. \qquad \square$$

Proof of Lemma A.2. The proof proceeds in three steps. First, Sobolev regularity is established by solving the equation on bounded domains and showing, through stochas-

tic representations, that the resulting solutions converge to the value function. Second, Sobolev embedding upgrades this regularity to local $C^{1,\gamma}$-smoothness. Finally, uniqueness is obtained from a Feynman–Kac representation and the contraction property inherited from the Bellman operator.

Fix $\theta \in \{n, r\}$ and a myopically optimal strategy α. By Proposition 5, $V_\theta \in C_b^0(\mathbb{R}^2)$. Under the strategy α, the reputation process (X_t^i, X_t^j) has infinitesimal generator

$$\mathcal{L}_\theta = \frac{\psi^2}{2}(\partial_{xx} + \partial_{yy}) + \mu_i \partial_x + \mu_j \partial_y,$$

where the drift coefficients μ_i and μ_j, given by (C.25) and (C.26), are bounded and Borel measurable. In particular, $\mathcal{L}_\theta$ is uniformly elliptic. I begin by deriving a resolvent representation of V_θ. By the fixed-point representation established in the proof of Proposition 5,

$$V_\theta(x, y) = \int_0^\infty 2\lambda e^{-2\lambda t} \left(\int_0^t e^{-\rho s} \pi \, ds + \frac{e^{-\rho t}}{2} \mathbb{E}_{x,y}[(\Upsilon_\theta + \Gamma_\theta)] \right) dt.$$

Applying Fubini's theorem and setting $\delta = \rho + 2\lambda$ gives

$$V_\theta(x, y) = \mathbb{E}_{x,y} \left[\int_0^\infty e^{-\delta s} \left[\pi + \lambda(\Upsilon_\theta + \Gamma_\theta) \right] ds \right]. \tag{C.30}$$

Hence, the HJB equation (A.1) is equivalently written as

$$\mathcal{L}_\theta V_\theta - \delta V_\theta = -\pi - \lambda(\Upsilon_\theta + \Gamma_\theta). \tag{C.31}$$

Claim 1: $V_\theta \in W^{2,p}_{\text{loc}}(\mathbb{R}^2)$ for every $p \in (1, \infty)$.

Because the Krylov–Itô formula used below requires an exponent greater than the dimension of the state space, first fix $p > 2$. Let $B_R \subset \mathbb{R}^2$ be an arbitrary open ball and define

$$\tau_B := \inf\{t \geq 0 : (X_t^i, X_t^j) \notin B_R\}.$$

The exit time τ_B is almost surely finite because the diffusion is nondegenerate and B_R is bounded. Applying the strong Markov property at τ_B to (C.30) gives

$$V_\theta(x, y) = \mathbb{E}_{x,y} \left[\int_0^{\tau_B} e^{-\delta s} \left[\pi + \lambda(\Upsilon_\theta + \Gamma_\theta) \right] ds + e^{-\delta \tau_B} V_\theta(X^i_{\tau_B}, X^j_{\tau_B}) \right] \tag{C.32}$$

The direct application of the Dirichlet existence theorem with boundary data V_θ is not available because V_θ is currently known only to be continuous. Choose instead a sequence $g_k \in C^\infty(\overline{B_R})$ such that

$$\sup_{\partial B_R} |g_k - V_\theta| \longrightarrow 0.$$

For each k, consider the Dirichlet problem

$$\begin{cases} \mathcal{L}_\theta \nu_k - \delta \nu_k = -\pi - \lambda(\Upsilon_\theta + \Gamma_\theta) & \text{in } B_R, \\ \nu_k = g_k & \text{on } \partial B_R. \end{cases} \tag{C.33}$$

The operator $\mathcal{L}_\theta - \delta$ is uniformly elliptic with constant principal coefficients, bounded first-order coefficients, and a strictly negative zeroth-order coefficient. Moreover, the right-hand side of (C.33) is bounded and therefore belongs to $L^p(B_R)$. Since $g_k \in W^{2,p}(B_R)$,Gilbarg and Trudinger (2001, Theorem 9.15) gives a unique strong solution

$$\nu_k \in W^{2,p}(B_R), \qquad \nu_k - g_k \in W_0^{1,p}(B_R).$$

Because $p > 2$, ν_k has a continuous representative on $\overline{B_R}$ whose boundary values agree with g_k. Applying the Krylov–Itô formula to $e^{-\delta(t\wedge\tau_B)}\nu_k(X^i_{t\wedge\tau_B}, X^j_{t\wedge\tau_B})$, using the usual localization to balls compactly contained in B_R, and then letting $t \to \infty$, gives

$$\nu_k(x,y) = \mathbb{E}_{x,y}\left[\int_0^{\tau_B} e^{-\delta s} \left[\pi + \lambda(\Upsilon_\theta + \Gamma_\theta)\right] ds + e^{-\delta\tau_B} g_k(X^i_{\tau_B}, X^j_{\tau_B}) \right]. \tag{C.34}$$

Comparing (C.34) with (C.32) yields

$$\begin{aligned} |\nu_k(x,y) - V_\theta(x,y)| &\le \mathbb{E}_{x,y}\left[e^{-\delta\tau_B} \left| g_k(X^i_{\tau_B}, X^j_{\tau_B}) - V_\theta(X^i_{\tau_B}, X^j_{\tau_B}) \right| \right] \\ &\le \sup_{\partial B_R} |g_k - V_\theta|. \end{aligned}$$

Consequently,

$$\nu_k \longrightarrow V_\theta \qquad \text{uniformly on } \overline{B_R}. \tag{C.35}$$

Applying the interior $W^{2,p}$ estimate on $B_{R/2} \Subset B_R$ gives

$$\|\nu_k\|_{W^{2,p}(B_{R/2})} \le C\left(\|\nu_k\|_{L^p(B_R)} + \left\|\pi + \lambda(\Upsilon_\theta + \Gamma_\theta)\right\|_{L^p(B_R)} \right),$$

where C is independent of k. The right-hand side is uniformly bounded in k by (C.35)

and the boundedness of V_θ, Υ_θ, and Γ_θ. Therefore, ν_k is bounded in $W^{2,p}(B_{R/2})$.

Since $1 < p < \infty$, the space $W^{2,p}(B_{R/2})$ is reflexive. Hence, a subsequence of ν_k converges weakly in $W^{2,p}(B_{R/2})$. Its weak limit must equal V_θ, because the full sequence converges uniformly to V_θ. It follows that $V_\theta \in W^{2,p}(B_{R/2})$.

Passing to the weak limit in (C.33) also shows that (C.31), and hence (A.1), holds almost everywhere on $B_{R/2}$. Since the center and radius of B_R and the exponent $p > 2$ were arbitrary,

$$V_\theta \in W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2) \qquad \text{for every } p > 2.$$

It remains to cover $p \in (1, 2]$. The preceding argument, applied with $p = 3$, gives $V_\theta \in W^{2,3}_{\mathrm{loc}}(\mathbb{R}^2)$. On every bounded ball B, finite measure implies $L^3(B) \subset L^p(B)$ for every $p \in (1, 2]$. Applying this inclusion to V_θ and to each of its weak derivatives of order at most two gives $W^{2,3}(B) \subset W^{2,p}(B)$, for $p \in (1, 2]$. Therefore, $V_\theta \in W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2)$ also for every $p \in (1, 2]$. Combining the two ranges proves Claim 1.

Claim 2: $V_\theta \in C^{1,\gamma}_{\mathrm{loc}}(\mathbb{R}^2)$ for every $\gamma \in (0, 1)$.

Fix $\gamma \in (0, 1)$ and choose $p > 2$ sufficiently large that

$$\gamma < 1 - \frac{2}{p}.$$

By the Sobolev embedding theorem in two dimensions (Gilbarg and Trudinger, 2001, Theorem 7.26), $W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2) \hookrightarrow C^{1,\gamma}_{\mathrm{loc}}(\mathbb{R}^2)$. Claim 1 therefore implies $V_\theta \in C^{1,\gamma}_{\mathrm{loc}}(\mathbb{R}^2)$ for every $\gamma \in (0, 1)$.

Claim 3: For every $p > 2$, V_θ is the unique bounded $W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2)$ strong solution of (A.1).

Fix $p > 2$ and suppose that $\widetilde{V}_\theta \in W^{2,p}_{\mathrm{loc}}(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$ is another strong solution of (A.1). Since $p > 2$, local Sobolev embedding gives a continuous representative of $\widetilde{V}_\theta$, and hence $\widetilde{V}_\theta \in C^0_b(\mathbb{R}^2)$. Applying the same localized Krylov–Itô argument used above to $\widetilde{V}_\theta$, and then letting the localization radius and time horizon tend to infinity, gives

$$\widetilde{V}_\theta(x, y) = \mathbb{E}_{x,y}\left[\int_0^\infty e^{-\delta s}\left[\pi + \lambda(\Upsilon_\theta + \Gamma_\theta)(X^i_s, X^j_s; \widetilde{V}_\theta)\right] ds\right].$$

Here the terminal term vanishes because $\widetilde{V}_\theta$ is bounded and $\delta > 0$. Reversing the Fubini calculation leading to (C.30), this representation is equivalent to $\widetilde{V}_\theta = \Phi^\alpha_\theta[\widetilde{V}_\theta]$. Proposition 5 establishes that Φ^α_θ is a contraction on $C^0_b(\mathbb{R}^2)$ and therefore has a unique

fixed point, namely V_θ. Since $\widetilde{V}_\theta$ is also a fixed point of Φ_θ^α, it follows that $\widetilde{V}_\theta = V_\theta$, proving uniqueness. □

Proof of Lemma A.3. Fix $\theta \in \{n, r\}$ and suppress the dependence on α. Set

$$\mathcal{U}_\theta := \{(x, y) \in \mathbb{R}^2 : V_\theta(x, y) = M_\theta(y)\}.$$

By Proposition 5, $\mathcal{C}_\theta(\alpha) = \{V_\theta > M_\theta\}$. Since V_θ and M_θ are continuous, $\mathcal{C}_\theta(\alpha)$ is open and $\partial\mathcal{C}_\theta(\alpha) \subseteq \mathcal{U}_\theta$. Therefore, it is enough to prove that $\mathcal{U}_\theta$ has Lebesgue measure zero.

Recall that

$$M_n(y) = \begin{cases} 0, & y < y_n, \\ (G_n + F_n)p(y) - F_n, & y > y_n, \end{cases}, \quad M_r(y) = \begin{cases} 0, & y < y_r, \\ (G_r + L_r)p(y) - L_r, & y > y_r, \end{cases}$$

The threshold line $\{y = y_\theta\}$ has two-dimensional Lebesgue measure zero. Thus, it is without loss to work separately on the two open half-planes $\{y < y_\theta\}$ and $\{y > y_\theta\}$, on each of which M_θ is smooth. By Lemma A.2, $V_\theta \in W^{2,p}_{\text{loc}}(\mathbb{R}^2)$ for every $p \in (1, \infty)$. Hence, $V_\theta - M_\theta \in W^{2,p}_{\text{loc}}$ on each of the two half-planes separated by $y = y_\theta$. Suppose that the contact set has positive measure in one of these half-planes. Since $V_\theta - M_\theta = 0$ on that set, Gilbarg and Trudinger (2001, Lemma 7.7), applied first to $V_\theta - M_\theta$ and then to each of its weak first derivatives, implies

$$\nabla(V_\theta - M_\theta) = 0 \quad \text{and} \quad \text{Hess}(V_\theta - M_\theta) = 0$$

almost everywhere on the contact set, where Hess denotes the matrix of second-order weak derivatives. Consequently, $\mathcal{L}_\theta V_\theta = \mathcal{L}_\theta M_\theta$ at almost every contact point. At such a point, the own-opportunity term $(M_\theta - V_\theta)^+$ vanishes. The HJB equation therefore reduces to the contact identity

$$0 = \pi + \mathcal{L}_\theta M_\theta(y) - \rho M_\theta(y) + \lambda \left[\Gamma_\theta(x, y; M_\theta) - M_\theta(y)\right]. \tag{C.36}$$

The expression on the right-hand side of (C.36) has only finitely many possible functional forms, at any state:

1. each partner-type continuation indicator is either zero or one;
2. whenever the partner stops, the action is either quit or the type-specific action;
3. the survival drift $\Lambda(y, x; \alpha)$ belongs to $\{-\lambda, 0, \lambda\}$.

Conditional on any of these finitely many configurations, the right-hand side of (C.36) is a real-analytic function of y.

The quitting region. Suppose $y < y_\theta$. Then

$$M_\theta(y) = 0, \qquad \mathcal{L}_\theta M_\theta(y) = 0,$$

and (C.36) reduces to

$$0 = \pi + \lambda \Gamma_\theta(x, y; 0). \tag{C.37}$$

For any configuration of the partner's stopping behavior, $\Gamma_\theta(x, y; 0)$ is affine in the posterior $p(y)$. Then (C.37) becomes

$$0 = \pi + \lambda \Big[\mathbb{1}\{(y,x) \notin \mathcal{C}_n\} b_\theta^n + p(y) \Big(\mathbb{1}\{(y,x) \notin \mathcal{C}_r\} b_\theta^r - \mathbb{1}\{(y,x) \notin \mathcal{C}_n\} b_\theta^n \Big) \Big].$$

where, for each partner type $\tilde{\theta}$, the indicators determine whether that type stops, and $b_\theta^{\tilde{\theta}}$ denotes the resulting terminal payoff to a type-θ player. Depending on the terminal action taken,

$$b_r^r \in \{0, G_r\}, \qquad b_r^n \in \{0, -L_r\},$$
$$b_n^r \in \{0, -L_n\}, \qquad b_n^n \in \{0, -L_n\}.$$

Thus, conditional on any fixed configuration (C.37) is an affine function of $p(y)$ and can be identically zero only if

$$\mathbb{1}\{(y,x) \notin \mathcal{C}_r\} b_\theta^r = \mathbb{1}\{(y,x) \notin \mathcal{C}_n\} b_\theta^n = -\frac{\pi}{\lambda}. \tag{C.38}$$

For $\theta = r$, the first term in (C.38) belongs to $\{0, G_r\}$ and the second to $\{0, -L_r\}$, so the equality is impossible. For $\theta = n$, however, the terms belong to $\{0, -L_n\}$. Thus, (C.38) can hold when both types stop through their type-specific terminal actions and $\pi/\lambda = L_n$. The payoff assumptions rules this out. Therefore, under every fixed configuration, the right-hand side of (C.37) is a nonzero affine function of $p(y)$. Since $p(\cdot)$ is strictly increasing, it can vanish at no more than one value of y.

The type-specific region. Suppose $y > y_\theta$. Using

$$p'(y) = p(y)(1 - p(y)); \quad p''(y) = p(y)(1 - p(y))(1 - 2p(y)); \quad \tanh(y/2) = 2p(y) - 1,$$

it follows that

$$\frac{\psi^2}{2} M_\theta''(y) + \frac{\psi^2}{2} \tanh(y/2) M_\theta'(y) = 0.$$

Consequently,

$$\mathcal{L}_\theta M_\theta(y) = \Lambda(y, x; \alpha) M'_\theta(y).$$

Since $\Lambda(y, x; \alpha) = \lambda\left[\mathbb{1}\{(y, x) \notin \mathcal{C}_n\} - \mathbb{1}\{(y, x) \notin \mathcal{C}_r\}\right]$, the term $\Lambda(y, x; \alpha) M'_\theta(y)$ cancels the terms quadratic in $p(y)$ that $\Gamma_\theta(x, y; M_\theta)$ generates. With this cancellation, (C.36) for the normative type reduces to

$$\begin{aligned} 0 = \pi + \rho F_n + \lambda \mathbb{1}\{(y, x) \notin \mathcal{C}_n\}(F_n + b_n^n) \\ + p(y)\Big\{ - \rho(G_n + F_n) + \lambda\big[\mathbb{1}\{(y, x) \notin \mathcal{C}_r\}(b_n^r - G_n) - \mathbb{1}\{(y, x) \notin \mathcal{C}_n\}(F_n + b_n^n)\big]\Big\}. \end{aligned}$$

This expression is affine in $p(y)$, so it vanishes at more than one posterior value only if it vanishes identically, which requires its constant term to be zero. Since $F_n + b_n^n \in \{F_n, F_n - L_n\}$, that term is strictly positive unless normative types whistleblows at (y, x). In that case it equals $\pi + \rho F_n - \lambda(L_n - F_n)$, which is nonzero by the genericity condition in footnote 3. The same steps for the rogue type give an affine expression with constant term $\pi + \rho L_r + \lambda \mathbb{1}\{(y, x) \notin \mathcal{C}_n\}(L_r + b_r^n)$, which is strictly positive because $L_r + b_r^n \in \{L_r, 0\}$. For both types, the expression therefore has at most one zero in $p(y)$, and since $p(\cdot)$ is strictly increasing, (C.36) has at most one solution y under each fixed configuration.

There are only finitely many configurations of the partner's continuation indicators and terminal actions. For each fixed configuration, the preceding arguments show that, in either half-plane, the contact identity can hold at no more than one value of y. Therefore, up to the null set on which either the HJB equation or the derivative identities fail, the contact set is contained in a finite union of horizontal lines, each has two-dimensional Lebesgue measure zero. Together with the nullity of the threshold line $\{y = y_\theta\}$, this implies

$$\mathrm{Leb}_2(\mathcal{U}_\theta) = 0.$$

Since $\partial\mathcal{C}_\theta \subseteq \mathcal{U}_\theta$, it follows that $\mathrm{Leb}_2\big(\partial\mathcal{C}_\theta\big) = 0$. □

Proof of Lemma A.4. Fix $\theta \in \{n, r\}$ and suppose $|V_\theta(x, y; \alpha) - M_\theta(y)| > \varepsilon$. There are two cases.

Case 1: $V_\theta(x, y; \alpha) - M_\theta(y) > \varepsilon$. Then $\varphi^*_\theta(x, y; \alpha) = c$, using $\|V_\theta(\cdot; \alpha) - V_\theta(\cdot; \widetilde{\alpha})\|_\infty \le \varepsilon$,

$$V_\theta(x, y; \widetilde{\alpha}) \;\ge\; V_\theta(x, y; \alpha) - \varepsilon \;>\; M_\theta(y),$$

so $\varphi^*_\theta(x, y; \widetilde{\alpha}) = c$.

Case 2: $V_\theta(x,y;\alpha) - M_\theta(y) < -\varepsilon$. Then $\varphi^*_\theta(x,y;\alpha) \neq c$, using $\|V_\theta(\cdot;\alpha) - V_\theta(\cdot;\widetilde{\alpha})\|_\infty \leq \varepsilon$,

$$V_\theta(x,y;\widetilde{\alpha}) \;\leq\; V_\theta(x,y;\alpha) + \varepsilon \;<\; M_\theta(y),$$

so $\varphi^*_\theta(x,y;\widetilde{\alpha}) \neq c$, and by myopic optimality $\varphi^*_\theta(x,y;\alpha) = \varphi^*_\theta(x,y;\widetilde{\alpha})$. □

Proof of Proposition 6. Throughout, the terminal payoffs, ρ, ψ, and $\pi/\lambda \in (0, L_n)$ are fixed. Constants denoted by C are finite, may increase between occurrences, and depend only on these fixed quantities. They are independent of λ, the conjectured strategy, and the initial state. Since $\pi/\lambda < L_n < L_r$, the restrictions $\lambda L_\theta > \pi$ hold throughout this family. For sufficiently small λ, $\rho G_\theta > \pi$ also holds, and the expression excluded in footnote 3 is strictly positive:

$$\pi + \rho F_n - \lambda(L_n - F_n) = \rho F_n + \lambda\left(\frac{\pi}{\lambda} - L_n + F_n\right) > 0.$$

For a bounded continuous pair $f = (f_n, f_r)$, let $\alpha[f]$ select collaboration where $f_\theta(x,y) > M_\theta(y)$ and $m_\theta(y)$ otherwise, using the selection in Proposition 5. Define

$$\mathcal{T}(f)_\theta := V_\theta(\cdot;\alpha[f]).$$

The value on the right is the one already defined in (3). A fixed point gives a symmetric equilibrium by Proposition 5. We construct a complete invariant set containing every equilibrium value and prove that $\mathcal{T}$ is a contraction on this set.

Uniform bounds and discounted occupation. For any myopically optimal conjecture α, the opportunity maps in (C.30) satisfy

$$\|\Upsilon_\theta(\cdot;V_\theta)\|_\infty + \|\Gamma_\theta(\cdot;V_\theta)\|_\infty \leq C + 2\|V_\theta\|_\infty.$$

Consequently, (C.30) gives $(\rho + 2\lambda)\|V_\theta\|_\infty \leq \pi + \lambda C + 2\lambda\|V_\theta\|_\infty$. Since π/λ is fixed,

$$\|V_\theta(\cdot;\alpha)\|_\infty \leq C\lambda. \tag{C.39}$$

No sign restriction is imposed. Applying the uniform interior $W^{2,4}$ estimates used in Lemma A.2 to (A.1), followed by Sobolev embedding, yields

$$\|\nabla V_\theta(\cdot;\alpha)\|_\infty \leq C\lambda. \tag{C.40}$$

Indeed, the value and the source in the equation are $O(\lambda)$ on every ball of fixed radius,

while the ellipticity and drift bounds in (1)–(2) are uniform for bounded λ.

We also use a discounted occupation estimate. Bounded measurable drift and constant nondegenerate diffusion give the uniform Gaussian bound

$$\mathbb{P}^{\alpha}_{x,y}\big((X^i_t, X^j_t) \in A\big) \le \int_A \frac{C}{t} \exp\!\left(-\frac{|(x,y)-(x',y')|^2}{Ct}\right) dx'\, dy', \qquad 0 < t \le 1. \tag{C.41}$$

If every vertical section of D has length at most $h \le 1$, integration first over that section bounds this probability by $C \min\{1, h/\sqrt{t}\}$. For $t \ge 1$, the Markov property gives Ch. Integration over time therefore yields

$$\sup_{x,y} \mathbb{E}^{\alpha}_{x,y}\left[\int_0^{\infty} e^{-\rho t} \mathbb{1}\{(X^i_t, X^j_t) \in D\}\, dt\right] \le Ch. \tag{C.42}$$

The same estimate holds for horizontal sections, for a finite union of sets with either kind of section bound, and with discount rate $\rho + 2\lambda$. It also holds when survival drift is zero.

A comparison value that permits quitting. Let $\mathcal{L}^0_\theta$ be the generator of (1)–(2) with survival drift omitted. Let V^0_θ be the unique bounded solution of

$$\begin{aligned}(\rho - \mathcal{L}^0_\theta) V^0_\theta(x,y) = \frac{\pi}{\lambda} + M_\theta(y) \\ + p(y) u^p(m_r(x);\theta) + (1-p(y)) u^p(m_n(x);\theta).\end{aligned} \tag{C.43}$$

This is the discounted resolvent under the signal-only dynamics. It is independent of λ because π/λ is fixed. The regularity argument in Lemma A.2 applies to this linear equation; in particular, V^0_θ has bounded continuous first derivatives.

Suppose $\max_\theta \|f_\theta\|_\infty \le C_0 \lambda$, with C_0 fixed, and write $V_\theta = V_\theta(\cdot; \alpha[f])$. The myopic rule and the positive right derivative of M_ϑ at y_ϑ imply that a partner can collaborate instead of taking a type-specific terminal action only inside

$$y_\vartheta < x < y_\vartheta + C\lambda, \qquad \vartheta \in \{n, r\}.$$

Below y_ϑ, her terminal action is quitting, so replacing it by collaboration changes the opportunity payoff only by $V_\theta = O(\lambda)$.

Subtract (C.43) from (A.1) divided by λ. Under the operator $\rho - \mathcal{L}^{\alpha[f]}_\theta$, the source

for $V_\theta/\lambda - V_\theta^0$ is bounded in absolute value by

$$C\lambda + C \sum_{\vartheta \in \{n,r\}} \mathbb{1}\{y_\vartheta < x < y_\vartheta + C\lambda\}. \tag{C.44}$$

Indeed, the own-opportunity discrepancy is $(M_\theta - V_\theta)^+ - M_\theta$, whose magnitude is at most $|V_\theta|$, and the partner-opportunity discrepancy has the bound just described. The remaining terms are $-V_\theta$ and $(\mathcal{L}_\theta^{\alpha[f]} - \mathcal{L}_\theta^0)V_\theta^0$. Both are $O(\lambda)$ by (C.39), $|\Lambda| \le \lambda$, and boundedness of ∇V_θ^0.

The bounded resolvent argument used in (C.30), together with (C.42), gives

$$\left\| \frac{V_\theta}{\lambda} - V_\theta^0 \right\|_\infty \le C\lambda. \tag{C.45}$$

In any ball of fixed radius, each strip in (C.44) has area $O(\lambda)$, so its L^4 norm is $O(\lambda^{1/4})$. Uniform interior $W^{2,4}$ estimates and Sobolev embedding then give

$$\left\| \nabla \left(\frac{V_\theta}{\lambda} - V_\theta^0 \right) \right\|_\infty \le C\lambda^{1/4}. \tag{C.46}$$

Choose C large enough and then λ sufficiently small. Let $\mathcal{K}_\lambda$ consist of bounded continuous pairs satisfying

$$\|f_\theta - \lambda V_\theta^0\|_\infty \le C\lambda^2, \tag{C.47}$$

$$\mathrm{Lip}(f_\theta - \lambda V_\theta^0) \le C\lambda^{5/4}. \tag{C.48}$$

The pair λV^0 belongs to this set, and the bounds are preserved by uniform limits, so it is nonempty and complete in the sup norm. Its elements satisfy a common bound $\|f_\theta\|_\infty \le C_0\lambda$. Equations (C.45)–(C.46) imply $\mathcal{T}(\mathcal{K}_\lambda) \subseteq \mathcal{K}_\lambda$.

Geometry of the quitting indifference sets. Under the signal-only dynamics the coordinates are independent, and $p(X_t^j)$ is a martingale. The partner-payoff contribution to V_θ^0 is therefore a weighted sum of one-dimensional discounted occupations of $[y_n, \infty)$ and $[y_r, \infty)$ by X^i. Each occupation has strictly positive derivative in the starting reputation x. The discounted expectation of $M_\theta(X_t^j)$ has strictly positive derivative in y at every finite y. The payoffs in Table 1 consequently give

$$\partial_x V_n^0(x,y) < 0, \qquad \partial_y V_r^0(x,y) > 0. \tag{C.49}$$

For the first inequality, both normative partner-payoff coefficients are negative. For

the second, the rogue partner-payoff contribution increases with $p(y)$, as does the own-opportunity contribution.

The tails make these crossing properties uniform near zero. As $x \to -\infty$, the partner-payoff contribution vanishes uniformly in y, leaving a positive limiting value at least $(\pi/\lambda)/\rho$. As $y \to -\infty$, uniformly in x,

$$V_\theta^0(x,y) \longrightarrow \frac{\pi/\lambda}{\rho} - L_\theta \, \mathbb{E}_x^0 \left[\int_0^\infty e^{-\rho t} \mathbb{1}\{X_t^i \geq y_n\} \, dt \,\middle|\, \theta_i = \theta \right]. \tag{C.50}$$

The right-hand side decreases from $(\pi/\lambda)/\rho > 0$ to $((\pi/\lambda) - L_\theta)/\rho < 0$, with strictly negative derivative at every finite x. The negative upper limit uses precisely $\pi/\lambda < L_n < L_r$.

For the normative type, as $x \to \infty$,

$$V_n^0(x,y) \longrightarrow \frac{(\pi/\lambda) - L_n}{\rho} + \mathbb{E}_y^0 \left[\int_0^\infty e^{-\rho t} M_n(X_t^j) \, dt \right]. \tag{C.51}$$

Its derivative in y is strictly positive on every compact interval. The relevant first derivatives also converge in these tail statements, by the one-dimensional occupation representations.

Here is an explicit finite partition of the near-zero sets on $y \leq y_\theta$. Choose $Y < \min(y_n, y_r)$ sufficiently negative. Equation (C.50) places all near-zero points with $y < Y$ in a fixed bounded interval $a < x < b$; there, $\partial_x V_\theta^0$ is bounded above by a negative constant. For the normative type, enlarge $[a,b]$ so that the remaining near-zero points either have $a \leq x \leq b$, where $\partial_x V_n^0$ has a uniform negative bound, or have $x > b$ and $Y \leq y \leq y_n$, where (C.51) gives a uniform positive bound on $\partial_y V_n^0$. For the rogue type, on $Y \leq y \leq y_r$ the derivative of the discounted expectation of M_r bounds $\partial_y V_r^0$ away from zero, uniformly in x. Outside these finitely many rectangular pieces, $|V_\theta^0|$ is bounded away from zero on $y \leq y_\theta$.

By (C.48), these coordinate crossing bounds persist for f_θ/λ when λ is small. Thus $\{|f_\theta| \leq \varepsilon\}$ near zero, restricted to $y < y_\theta$, is a finite union of pieces whose horizontal or vertical sections have length at most $C\varepsilon/\lambda$.

Disagreement between nearby strategies. Take $f, g \in \mathcal{K}_\lambda$ and put $\varepsilon := \max_\theta \|f_\theta - g_\theta\|_\infty$. By (C.47), $\varepsilon \leq 2C\lambda^2$. If $\alpha_\theta[f] \neq \alpha_\theta[g]$, their comparisons with M_θ have opposite signs, so $|f_\theta - M_\theta| \leq \varepsilon$. This is the sign argument in Lemma A.4. Split the disagreement set into D_θ^q on $y < y_\theta$ and D_θ^+ on $y > y_\theta$; the threshold line is null and does not affect

occupation or payoffs.

On D_θ^q the preceding argument gives section lengths at most $C\varepsilon/\lambda$ in finitely many pieces. On D_θ^+, $M_\theta(y) \le C\lambda$, so $y_\theta < y < y_\theta + C\lambda$. In this strip M_θ' is bounded below by a positive constant, whereas $\mathrm{Lip}(f_\theta) \le C\lambda$. Thus every vertical section of D_θ^+ has length at most $C\varepsilon$. By (C.42),

$$\sup_{x,y} \mathbb{E}_{x,y}^{\alpha} \left[\int_0^\infty e^{-(\rho+2\lambda)t} \mathbb{1}_{D_\theta^q}(X_t^i, X_t^j)\, dt \right] \le C\frac{\varepsilon}{\lambda}, \tag{C.52}$$

$$\sup_{x,y} \mathbb{E}_{x,y}^{\alpha} \left[\int_0^\infty e^{-(\rho+2\lambda)t} \mathbb{1}_{D_\theta^+}(X_t^i, X_t^j)\, dt \right] \le C\varepsilon. \tag{C.53}$$

The same bounds hold for the transposed sets, which exchange the players' coordinates, and are uniform over the conjectured strategy generating the diffusion.

Contraction. Write $\alpha = \alpha[f]$, $\widetilde{\alpha} = \alpha[g]$, $V_\theta = \mathcal{T}(f)_\theta$, and $\widetilde{V}_\theta = \mathcal{T}(g)_\theta$. Subtracting their equations (C.31), with the generator induced by α on the left, gives

$$\begin{aligned}
&(\rho + 2\lambda - \mathcal{L}_\theta^\alpha)(V_\theta - \widetilde{V}_\theta) \\
&= \lambda\big[\Upsilon_\theta(V_\theta) - \Upsilon_\theta(\widetilde{V}_\theta)\big] \\
&\quad + \lambda\big[\Gamma_\theta^\alpha(V_\theta) - \Gamma_\theta^\alpha(\widetilde{V}_\theta)\big] \\
&\quad + \lambda\big[\Gamma_\theta^\alpha(\widetilde{V}_\theta) - \Gamma_\theta^{\widetilde{\alpha}}(\widetilde{V}_\theta)\big] \\
&\quad + (\mathcal{L}_\theta^\alpha - \mathcal{L}_\theta^{\widetilde{\alpha}})\widetilde{V}_\theta.
\end{aligned} \tag{C.54}$$

The strategy superscripts only indicate the dependence already present in Γ_θ and $\mathcal{L}_\theta$. The bounded resolvent argument in Lemma A.2 applies to this difference equation. Both opportunity maps are one-Lipschitz in the continuation value, as shown in the proof of Proposition 5. The first two terms contribute at most

$$\frac{2\lambda}{\rho + 2\lambda} \|V_\theta - \widetilde{V}_\theta\|_\infty.$$

On a transposed quitting disagreement set, the partner chooses between quitting, which pays zero, and collaboration, which leaves $\widetilde{V}_\theta = O(\lambda)$. On a transposed type-specific disagreement set, the payoff change is bounded by a constant. Hence

$$\left| \Gamma_\theta^\alpha(\widetilde{V}_\theta) - \Gamma_\theta^{\widetilde{\alpha}}(\widetilde{V}_\theta) \right| \le C\lambda \sum_\vartheta \mathbb{1}_{(D_\vartheta^q)^\top} + C \sum_\vartheta \mathbb{1}_{(D_\vartheta^+)^\top}.$$

After multiplication by λ, equations (C.52)–(C.53) bound the discounted contribution by $C\lambda\varepsilon$.

The generators differ only through survival drift, by (1)–(2). Its change is bounded by λ times disagreement indicators in the relevant coordinate order. Together with (C.40), this gives

$$\left|(\mathcal{L}_\theta^\alpha - \mathcal{L}_\theta^{\widetilde{\alpha}})\widetilde{V}_\theta\right| \leq C\lambda^2 \sum_\vartheta \left(\mathbb{1}_{D_\vartheta^q} + \mathbb{1}_{D_\vartheta^+} + \mathbb{1}_{(D_\vartheta^q)^\top} + \mathbb{1}_{(D_\vartheta^+)^\top}\right).$$

The same occupation estimates bound this contribution by $C\lambda\varepsilon$. Combining the bounds and absorbing the direct continuation-value terms yields

$$\|\mathcal{T}(f) - \mathcal{T}(g)\|_\infty \leq C\lambda\|f - g\|_\infty. \tag{C.55}$$

For sufficiently small λ, $\mathcal{T}$ is a contraction on the complete invariant set $\mathcal{K}_\lambda$. Banach's theorem gives a unique fixed point, and Proposition 5 identifies its induced strategy as a symmetric Markov perfect equilibrium.

Finally, every equilibrium value satisfies (C.39)–(C.40). The verification argument for Proposition 5 implies that an equilibrium rule agrees almost everywhere with an optimal opportunity choice. Away from indifference it therefore agrees almost everywhere with the selected rule comparing its value with M_θ and taking m_θ on stopping. Lemma A.3 makes the indifference sets null as well. Replace the equilibrium rules on these null sets by the pointwise selection in Proposition 5. By Lemma A.1, this does not change occupation times, the law at positive Poisson opportunities, or the equilibrium values. The approximation argument now applies and puts every equilibrium value in $\mathcal{K}_\lambda$. Every equilibrium must consequently have the fixed-point value just constructed.

That value determines a unique pointwise selected strategy. Any remaining changes to an equilibrium rule are confined to null sets and leave reputation dynamics and the equilibrium stopping and payoff laws unchanged. This establishes all the uniqueness claims.

□

Proof of Proposition 7, part (i). Fix a symmetric Markov equilibrium and assume, in addition, $F_n \leq L_n$. Suppress the equilibrium argument in the values. Use $\delta = \rho + 2\lambda$

from (C.31) and set $\eta = (1 + 8\delta/\psi^2)^{-1/2}$. Define

$$\begin{aligned} B_0(y) &= \frac{\pi - \lambda L_n + \lambda M_n(y)}{\delta}, \\ B(y) &= B_0(y) + \frac{\lambda\eta G_n p(y)}{\delta} \exp\left[-\frac{y - y_n}{2} - \frac{|y - y_n|}{2\eta}\right]. \end{aligned} \tag{C.56}$$

Direct differentiation on either side of y_n verifies the comparison equation below. The correction cancels the first-derivative jump of B_0 at y_n; the continuous right-hand side then makes B'' continuous. Thus B is bounded and C^2, with

$$\delta B - \frac{\psi^2}{2}\big(B'' + \tanh(y/2)B'\big) = \pi - \lambda L_n + \lambda M_n(y). \tag{C.57}$$

The correction in (C.56) attains its maximum at y_n. Since $M_n \geq 0$, it follows that

$$\begin{aligned} B(y) - M_n(y) &\leq \frac{\lambda}{\delta}\big(\bar{L}_n^\infty - L_n\big), \\ B(y_n) &= \frac{\lambda}{\delta}\big(\bar{L}_n^\infty - L_n\big), \qquad \bar{L}_n^\infty = \frac{\pi}{\lambda} + \eta\frac{G_n F_n}{G_n + F_n}. \end{aligned} \tag{C.58}$$

Openness. Suppose $L_n < \bar{L}_n^\infty$. The correction satisfies

$$\frac{B - B_0}{1 - p} \leq \frac{\lambda\eta F_n}{\delta} \quad (y \leq y_n), \qquad \frac{B - B_0}{p} \leq \frac{\lambda\eta G_n}{\delta} \quad (y \geq y_n).$$

Differentiating (C.56) and using these bounds gives

$$\begin{aligned} (1 - p)(B + L_n) - B' &\geq \frac{(1 - p)[\pi + (\rho + \lambda)L_n - \lambda F_n]}{\delta} > 0, \\ p(B + L_n) + B' &\geq \frac{p[\pi + (\rho + \lambda)L_n]}{\delta} > 0. \end{aligned} \tag{C.59}$$

The first strict inequality uses $\pi + (\rho + \lambda)L_n - \lambda F_n \geq \pi + \rho L_n > 0$. Continuity supplies both bounds at y_n.

Equation (C.57) treats both partner types as taking their harmful terminal actions. By (A.1) and (C.26), replacing a normative partner's action by collaboration changes the Bellman residual at B by $\lambda[(1-p)(B+L_n) - B']$; replacing a rogue partner's action by collaboration changes it by $\lambda[p(B + L_n) + B']$. Quitting replaces the loss $-L_n$ with zero, and own optimization adds $\lambda(B - M_n)^+$. All these changes are nonnegative by (C.59), so B is a subsolution of (A.1). Lemma A.2 and bounded discounted comparison give $V_n(x, y) \geq B(y)$: both opportunity maps are nondecreasing and one-Lipschitz in the continuation value, so their linearization leaves a discount coefficient at least $\rho > 0$.

Evaluating at y_n and applying (C.58) yields

$$V_n(x, y_n) \geq \frac{\lambda}{\delta}\big(\bar{L}_n^{\infty} - L_n\big) > 0 = M_n(y_n) \qquad \text{for every } x \in \mathbb{R}.$$

Proposition 5 therefore gives openness on $\mathbb{R}$, with the same partner reputation y_n for every x.

Fear. Suppose $L_n > \bar{L}_n^{\infty}$. Every terminal payoff to type θ is at most G_θ. Since $\rho G_\theta > \pi$, paying G_θ at the first opportunity of either player gives the uniform upper bound

$$V_\theta(x, y) \leq \frac{\pi + 2\lambda G_\theta}{\rho + 2\lambda} < G_\theta.$$

Together with $M_\theta(x) \to G_\theta$ and Proposition 5, this gives a finite $\bar{x}$ such that, whenever $x > \bar{x}$, a normative partner whistleblows and a rogue partner schemes, regardless of y. Thus $\Gamma_n = -L_n$ and $\Lambda(y, x; \alpha) = 0$ there. Subtracting (C.57) from (A.1), using (C.26), gives

$$(\delta - \mathcal{L}_n)(V_n - B) = \lambda(V_n - M_n)^+, \qquad x > \bar{x}. \tag{C.60}$$

Let $\tau_{\bar{x}} = \inf\{t \geq 0 : X_t^i \leq \bar{x}\}$. The bounded drift in (1) makes the probability of reaching $\bar{x}$ over any fixed finite horizon vanish uniformly in y as $x \to \infty$. Splitting the discounted expectation at that horizon therefore gives, for every $q > 0$,

$$\sup_y \mathbb{E}_{x,y}\big[e^{-q\tau_{\bar{x}}} \mathbb{1}\{\tau_{\bar{x}} < \infty\}\big] \longrightarrow 0 \qquad (x \to \infty). \tag{C.61}$$

By (C.58), $B - M_n$ is uniformly strictly negative, so $(V_n - M_n)^+ \leq (V_n - B)^+$. Positive-part comparison in (C.60), stopped at $\tau_{\bar{x}}$, then gives a finite constant C with

$$(V_n - B)^+(x, y) \leq C\mathbb{E}_{x,y}\big[e^{-(\rho+\lambda)\tau_{\bar{x}}} \mathbb{1}\{\tau_{\bar{x}} < \infty\}\big]. \tag{C.62}$$

Combining (C.62), (C.61), and (C.58) gives

$$V_n(x, y) - M_n(y) \leq -\frac{\lambda}{\delta}\big(L_n - \bar{L}_n^{\infty}\big) + o(1) < 0$$

for all sufficiently high x, uniformly in y. Proposition 5 therefore gives fear on an upper half-line, as claimed. □

Proof of Proposition 8. For two normative players, the partnership survives forever with strictly positive probability. Thus, for every $(x, y) \in \mathbb{R}^2$,

$$\mathbb{P}\big(\tau < \infty \,\big|\, \theta_i = \theta_j = n,\, X_0^i = x, X_0^j = y\big) < 1.$$

To prove this claim, first establish that both normative players strictly prefer collaboration when both reputations are sufficiently low. Choose

$$x_2 < \min\left\{\log\frac{F_n}{G_n}, \log\frac{L_r}{G_r}\right\}.$$

Whenever a player's reputation lies below x_2, either partner type strictly prefers quitting to her type-specific terminal action. Consequently, any partner exit while that player's reputation remains below x_2 pays her zero.

Consider a normative player who quits at her first opportunity, T_1^i. This strategy collects flow payoffs at least until the first opportunity of either player, giving an expected discounted flow of at least $\pi/(\rho + 2\lambda)$. A negative terminal payoff can occur only if her reputation reaches x_2 before T_1^i. Her reputation drift is bounded above by $\lambda - \psi^2/2$, so

$$\begin{aligned} V_n(x,y) \geq{}& \frac{\pi}{\rho + 2\lambda} \\ &- L_n\,\mathbb{P}\left(\sup_{0\leq s\leq T_1^i}\left\{\left(\lambda - \frac{\psi^2}{2}\right)s + \psi B_s^i\right\} \geq x_2 - x\right), \qquad x < x_2. \end{aligned}$$

This bound holds uniformly in the partner's reputation and actual type. Since $T_1^i < \infty$ almost surely, the running maximum inside this probability is finite almost surely. Its tail therefore vanishes as $x \to -\infty$. Hence there exists $x_1 < x_2$ such that

$$V_n(x,y) > 0 \qquad \text{for every } x < x_1 \text{ and every } y \in \mathbb{R}.$$

When both reputations lie below x_1, the normative terminal rewards satisfy $M_n(x) = M_n(y) = 0$. Both normative players therefore strictly prefer collaboration throughout

$$(-\infty, x_1) \times (-\infty, x_1).$$

Inside this region, normative collaboration implies

$$\Lambda(x, y; \alpha) = \lambda\left[\mathbb{1}\{\alpha_r(x,y) = c\} - 1\right] \leq 0,$$

and likewise $\Lambda(y, x; \alpha) \leq 0$. Conditional on both actual types being normative, each reputation therefore has drift at most $-\psi^2/2$ until the pair leaves the region.

Starting from $x, y < x_1$, compare the two reputations with the independent processes

$$x - \frac{\psi^2}{2}t + \psi B_t^i, \qquad y - \frac{\psi^2}{2}t + \psi B_t^j.$$

If neither comparison process reaches x_1, neither reputation can leave the collaboration region. Both players then collaborate at every opportunity, so $\tau = \infty$.

The probability that a Brownian motion with drift $-\psi^2/2$ and volatility ψ, starting at $x < x_1$, ever reaches x_1 is e^{x-x_1}. Independence consequently gives

$$\mathbb{P}\big(\tau = \infty \,\big|\, \theta_i = \theta_j = n,\, X_0^i = x, X_0^j = y\big) \geq (1 - e^{x-x_1})(1 - e^{y-x_1}) > 0, \qquad x, y < x_1.$$

Finally, consider any finite initial profile. Fix a deterministic time $t > 0$. The global drift bound gives

$$X_t^i \leq x + \left(\lambda - \frac{\psi^2}{2}\right) t + \psi B_t^i,$$
$$X_t^j \leq y + \left(\lambda - \frac{\psi^2}{2}\right) t + \psi B_t^j.$$

There is positive probability that both Brownian endpoints are sufficiently negative to place both reputations below $x_1 - 1$. Independently, neither player receives an opportunity before t with probability $e^{-2\lambda t} > 0$, so the partnership remains active until then. From that state, the preceding comparison gives a probability of permanent collaboration of at least $(1 - e^{-1})^2 > 0$.

Therefore every normative–normative partnership has strictly positive probability of surviving forever, and

$$\mathbb{P}\big(\tau < \infty \,\big|\, \theta_i = \theta_j = n,\, X_0^i = x, X_0^j = y\big) < 1.$$

□

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